



Smart Highway Speed Violation Prediction Using Machine Learning and RFID Technology

Project Proposal Report

Project ID: 25-26J-314

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August 2025

Declaration

I hereby declare that this proposal is my own work and does not incorporate, without proper acknowledgment, any material previously submitted for a degree or diploma in any other university or institute of higher learning. I also declare that no part of this work has been previously published or written by another person, except where due reference is made in the text.

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Abstraction

Over-speeding is one of the most critical causes of road traffic accidents and fatalities both globally and in Sri Lanka. While existing enforcement mechanisms, such as speed guns and static speed cameras, serve as deterrents, they remain reactive, limited in coverage, and lack predictive capabilities. The inability to anticipate high-risk violations or profile repeat offenders has created a gap in proactive traffic enforcement.

This research focuses on developing the machine learning component of an integrated RFID and IoT-based smart speed monitoring system. RFID readers installed on highways detect speed violations by reading embedded RFID chips in driving licenses. The violation data collected will then be processed by machine learning algorithms to identify patterns in traffic violations, enabling predictive analysis of future risks. Specifically, the proposed ML model will analyze violation histories to forecast high-risk vehicle types, times of day, and road segments.

The system introduces a novel data-driven layer of prediction and visualization. In addition to real-time violation detection, the research contributes a predictive reporting module that presents authorities with interactive charts and insights, empowering them to allocate enforcement resources more strategically and launch targeted awareness campaigns. The inclusion of image processing to classify vehicle types further enhances the model's accuracy.

The novelty of this work lies in bridging the enforcement–prediction gap by extending traditional speed monitoring with machine learning–powered forecasting and reporting. This component will transform traffic law enforcement in Sri Lanka from a reactive practice into a proactive, evidence-based approach to saving lives and enhancing road safety.

Table of Contents

Declaration	2
Abstraction	3
List of Figure	5
1. INTRODUCTION	6
1.1. Background & Literature Survey	6
1.2. Research Gap	8
1.3. Research Problem.....	8
2. OBJECTIVES.....	10
2.1 Main Objective.....	10
2.2 Specific Objectives.....	10
3. METHODOLOGY.....	11
3.1 Project Overview.....	11
3.2. System diagram.....	12
3.2 Individual Component	13
4. RESEARCH & DEVELOPMENT OVERVIEW	15
5. BUDGET AND BUDGET JUSTIFICATION	18
References	22

List of Figure

Figure 1 System diagram.....	12
Figure 2 Budget plan.....	19
Figure 3 Gantt chart	20
Figure 4 WBS diagram.....	21

1. INTRODUCTION

Rapid urbanization has compounded on-street parking and roadside activity including loading/unloading, ride-hailing stops, and short-term waiting, resulting in transient congestions detrimental to traffic flow, safety, and emissions, in Sri Lanka. Conventional parking management strategies (static signage, manual enforcement and train surveys) fail to capture the variability and short-term nature of the flows that are truly disruptive to capacity at actual times and places. Recent advances in computer vision, embedded systems now allow real-time, uninterrupted monitoring of curb side action and how it directly affects traffic performance.

This paper presents a Dynamic Roadside Behavior Analysis system based on a custom YOLOv5 detector and Deep SORT multi-object tracker to identify parked vehicles and roadside activity; measure the parking duration per-vehicle classify and alert of illegal parking in no parking zones; and measure the impact of roadside behavior on traffic density The system can be connected to Arduino-based LEDs and buzzers to provide on-site alerts and to a central dashboard, signal control interfaces to respond to network level situations. Aimed at the needs of heterogeneous and varied parking patterns of Sri Lanka, the solution is expected to be deployed in practical and real conditions along busy streets and in close proximity to activity clusters (schools, markets, bus stops, and hospitals) in order to permit operation based on data and support real-time traffic management.

The contributions expected are four-fold a powerful curbside detection and tracking VIS pipeline tailored to the traffic realities in Sri Lanka real-time detection of parking duration and roadside behaviors classification (e.g., loading, passenger drop-off) quantitative relation between roadside behaviors and traffic flow measures and integrated alerting and control framework that reduces the enforcement latency and proactive signal designs support.

1.1. Background & Literature Survey

Speeding is a major contributor to traffic accidents globally. According to the World Health Organization (2024), nearly 1.3 million people die every year due to road traffic crashes, with over speeding identified as a leading factor. Sri Lanka faces a growing road safety challenge due to rapid urbanization, increased private vehicle ownership, and limited automated enforcement mechanisms. Current practices rely heavily on speed guns and static cameras, which are reactive, prone to human error, and cover only limited road segments.

Internationally, RFID and IoT-based systems have been employed for toll collection, congestion monitoring, and fleet management. For example, India's Fastag system automates toll collection using RFID, while Singapore has integrated smart road pricing systems. Research from China and the U.S. demonstrates that when IoT sensors are combined with predictive analytics, authorities can anticipate congestion and violations with significant accuracy. However, predictive enforcement of speed violations using RFID data integrated with machine learning remains underexplored.

Machine learning techniques such as Random Forest, Gradient Boosting, and Neural Networks have shown strong results in traffic violation prediction, anomaly detection, and risk profiling. Yet, few implementations target Sri Lanka's unique traffic patterns, which involve heterogeneous vehicle types, irregular driving behavior, and limited enforcement resources.

1.2. Research Gap

Despite advancements in traffic monitoring technologies, Sri Lanka still lacks predictive enforcement tools that can proactively identify high-risk drivers, timeframes, and locations. Current RFID use is limited to toll collection, and while IoT research in Sri Lanka has explored real-time monitoring, predictive analytics remains underutilized. The absence of integrated visualization platforms also restricts enforcement agencies from leveraging insights for strategic planning.

This research addresses the gap by integrating RFID violation data with machine learning techniques to produce predictive insights, allowing authorities to transition from reactive to proactive traffic enforcement.

1.3. Research Problem

How can RFID-based speeding violation data be effectively transformed through machine learning models to predict future high-risk speeding patterns—categorized by driver, vehicle type, location, and time—thereby enabling authorities in Sri Lanka to adopt proactive traffic enforcement strategies?

Expanded Description:

1. Data Acquisition and Collection

- Modern highways in Sri Lanka can collect vehicle information in real-time using **RFID tags** installed in vehicles and RFID readers along highways.
- Each record captures critical attributes such as **vehicle ID, vehicle type, speed, location, date, and time**.
- This research aims to utilize this **large-scale IoT-generated dataset** for predictive analysis.

2. Data Transformation and Preprocessing

- Raw RFID data is often **noisy, incomplete, or inconsistent**.
- The study focuses on **cleaning, normalizing, and structuring the data** so that machine learning models can process it effectively.
- Features such as **driver behavior patterns, vehicle type, traffic density, and time of day** will be extracted to enhance prediction accuracy.

3. Predictive Modeling Using Machine Learning

- The core challenge is to **build accurate machine learning models** that can identify

patterns in historical speeding data.

- Models aim to predict **high-risk speeding incidents**, considering multiple factors:
 - **Driver behavior patterns** (frequent offenders, risk-prone behavior)
 - **Vehicle type** (heavy vehicles vs. light vehicles)
 - **Location hotspots** (specific stretches of highways with frequent violations)
 - **Time patterns** (peak hours, weekends, or seasonal variations)

4. Categorization and Risk Profiling

- The study will categorize predicted violations by **driver, vehicle, location, and time**, creating a **risk profile** for each category.
- These profiles help in identifying **potential future offenders or risky areas**, allowing authorities to focus enforcement efforts efficiently.

5. Proactive Traffic Management

- By predicting **where and when speeding violations are likely to occur**, traffic authorities can implement **preventive strategies** rather than reactive measures.
- Examples include:
 - Targeted **patrolling** in high-risk zones
 - Deployment of **dynamic speed control measures**
 - **Awareness campaigns** for drivers in risk-prone areas

6. Impact on Road Safety in Sri Lanka

- Implementing predictive analytics will help **reduce road accidents, minimize fatalities, and enhance traffic law enforcement efficiency**.
- The research contributes to the **adoption of smart highway technologies** in Sri Lanka, promoting safer and more intelligent traffic systems.

2. OBJECTIVES

The ultimate objective of this study would be to generate a real-time roadside behavior monitoring system that uses computer vision and embedded devices to identify, track, and analyze parking activities, with the view to discerning how they affect the traffic flow of Sri Lanka due to its mixed-traffic situations.

2.1 Main Objective

To design and develop a smart highway speed monitoring and predictive enforcement system that integrates RFID-based driver identification, IoT-enabled speed tracking, and machine learning to detect violations, notify authorities, and forecast high-risk driver and road profiles.

2.2 Specific Objectives

- Develop a machine learning model for predictive risk analysis using RFID violation data.
- Apply image processing techniques for accurate vehicle type classification.
- Design a visualization dashboard for predictive insights.
- Validate the system with simulated and secondary traffic datasets.
- Provide actionable recommendations for enforcement agencies

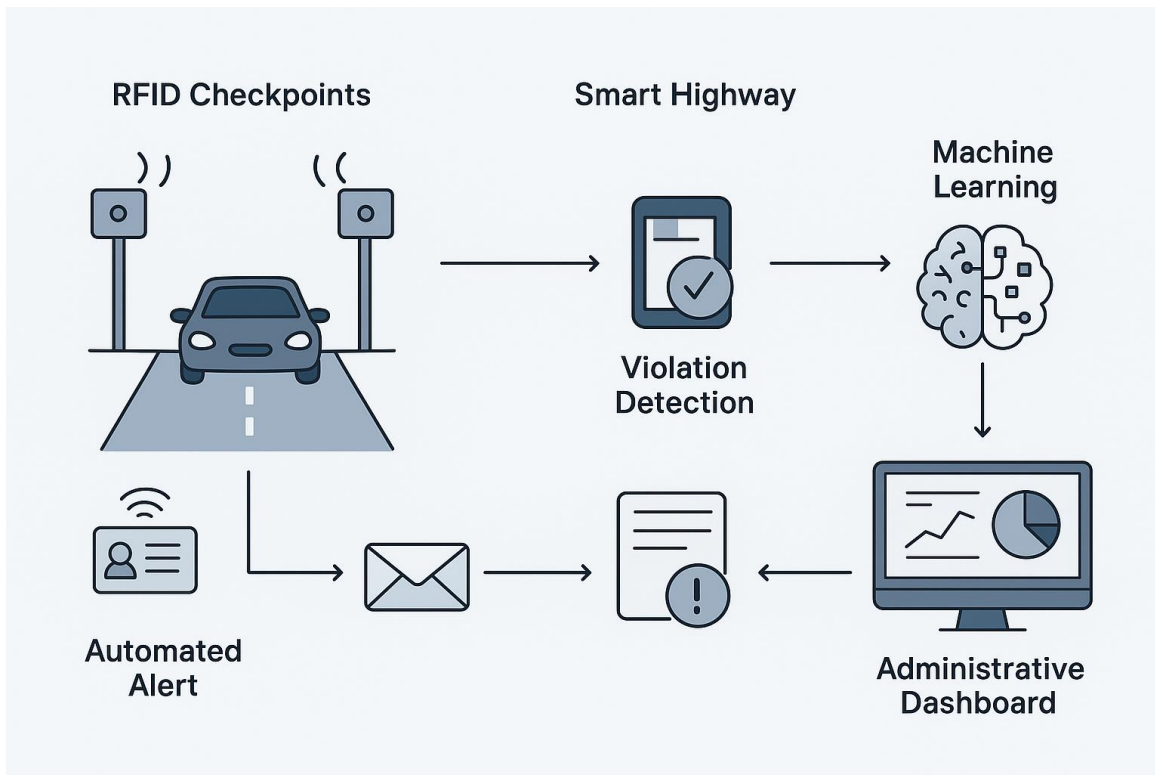
3. METHODOLOGY

3.1 Project Overview

The proposed methodology follows a data-driven machine learning pipeline. Data will be collected from RFID readers placed at highway entry and exit points. This dataset will be cleaned, preprocessed, and augmented with time, location, and vehicle attributes. Machine learning algorithms will be trained to identify patterns and forecast future violations. An image processing module will classify vehicles, while a visualization dashboard will present predictive insights to traffic authorities.

3.2 System Diagram

Figure 1 illustrates the system architecture for the proposed model.



3.3 Individual Components

Data Preprocessing: Cleaning, feature engineering, and normalization of violation datasets.

Machine Learning Model: Training predictive models using Random Forest, Gradient Boosting, and Neural Networks.

Image Processing: Vehicle classification using Convolutional Neural Networks (CNNs).

Visualization Dashboard: Displaying predictive insights with charts and reports.

Integration: Linking all components into a single decision-support system.

The methodology adopts four major stages:

Data Collection & Preprocessing-Collect Road side video data sets of major urban corridors in Sri Lanka and in varied conditions (day/night, weather, traffic density). Label data to train YOLOv5 to detect parked cars, roadside stops and activities like loading/unloading and passenger drop-offs.

Model Development and Training - Train a Yorkshire Tasty YOLOv5 object detection model of parked vehicles and activity recognition optimized with the Sri Lankan traffic in mind. Apply Deep SORT multi-object tracking to count the parking time of each vehicle.

System Integration -Integrate illegal parking detection with no-parking geo-fenced zones. Connect Arduino-based instruments to the vision system (LED, buzzers) that can provide immediate indication of a violation. Build a central data visualization, enforcement and decision-making dashboard, which has connectivity to traffic signal control systems to support adaptive operation.

Conduct field tests on a sub-set of Sri Lankan roads to test model accuracy, reliability of the estimation parking duration, effectiveness of illegal parking detection and overall effects the model has on the analysis of traffic flow. Performance gained will be benchmarked against the performance of a baseline against traffic without the system

3.1 Individual Component

3.1.1 Data Preprocessing

- **Cleaning and Preparation:** Raw RFID and IoT datasets collected from highways will often contain missing values, inconsistencies, and noise. This stage involves **removing invalid entries, handling missing data, and correcting inconsistencies**.
- **Feature Engineering:** Important features such as **vehicle type, driver ID, speed, location, time of day, and day of the week** will be derived to improve model performance.
- **Normalization:** All numerical data, such as vehicle speed or time intervals, will be **scaled**

and normalized to ensure uniformity and compatibility with machine learning algorithms.

- **Categorization:** Categorical data such as vehicle type and location will be **encoded appropriately** for machine learning models.

3.3.2 Machine Learning Module

- **Model Selection:** Predictive models including **Random Forest, Gradient Boosting, and Neural Networks** will be trained to forecast high-risk speeding events.
- **Training and Validation:** Historical RFID violation data will be split into **training, validation, and test sets** to optimize model performance.
- **Prediction Objectives:** Models will aim to **predict future speeding violations**, categorize high-risk drivers, vehicle types, locations, and time periods.
- **Performance Metrics:** Accuracy, precision, recall, and F1-score will be measured to **evaluate and compare model performance**.

3.3.3 Image Processing (Optional/Enhanced Component)

- **Vehicle Classification:** Using **Convolutional Neural Networks (CNNs)**, vehicles may be classified in visual datasets (if integrated with camera data) to enhance driver and vehicle-type prediction.
- **Activity Detection:** Detect abnormal or risky behaviors, such as sudden lane changes or overtaking at high speeds, to complement RFID-based predictions.

3.3.4 Visualization Dashboard

- **Data Representation:** Display predictive insights, violation trends, and risk patterns using **interactive charts, heatmaps, and reports**.
- **User Accessibility:** Dashboard will provide an intuitive interface for **traffic authorities and policymakers** to monitor and analyze violations.
- **Real-Time Updates:** Optionally, the dashboard can show **live alerts** for high-risk violations

based on predictive analytics.

3.3.5 Integration

- **System Integration:** All modules (data preprocessing, machine learning, and visualization) will be **linked into a single decision-support system**.
- **Decision Support:** Traffic authorities can **leverage predictions for proactive enforcement**, dynamic traffic management, and policy planning.
- **Scalability:** The integrated system will be designed to **scale across multiple highways and urban corridors** in Sri Lanka.

4. Research & Development Overview

The research and development phase of this project focuses on **leveraging RFID and IoT data to predict speeding violations on Sri Lankan highways**. The aim is to design a system that not only identifies high-risk drivers and vehicles but also provides actionable insights for **proactive traffic management and enforcement**. This section elaborates on data sources, anticipated benefits, scope, deliverables, and constraints.

4.1 Sources of Data

To ensure the development of a robust and accurate predictive system, the project will utilize **multiple data sources**:

1. **Simulated RFID Datasets**
 - Since real-world RFID violation data may be limited, **synthetic datasets** will be generated to simulate vehicle movements, speeds, and violation patterns.
 - Simulated data allows the team to **test and fine-tune machine learning models** under controlled conditions while covering a wide range of scenarios.
2. **Secondary Datasets from Global Traffic Repositories**
 - Publicly available traffic violation datasets and IoT vehicle tracking records from global sources will be used for **benchmarking and model validation**.
 - These datasets help in understanding patterns and trends that can be adapted to the **Sri Lankan traffic context**.
3. **Annotated Image Datasets**
 - To enhance vehicle classification and driver behavior prediction, **annotated datasets of different vehicle types** (cars, buses, motorcycles, three-wheelers) will be used.
 - These images support the development of **computer vision modules** and can be integrated with RFID-based predictions for a more comprehensive system.
4. **WHO and Government Traffic Statistics**
 - Baseline accident, traffic violation, and vehicle population statistics will be sourced from **World Health Organization (WHO) reports and Sri Lankan government transport data**.
 - These statistics provide **contextual benchmarks** and help in evaluating the effectiveness of the predictive system.

4.2 Anticipated Benefits

The project is expected to provide **tangible improvements in traffic safety and management**, including:

- **Proactive Traffic Enforcement**
 - By predicting high-risk speeding events, authorities can take **preventive measures** rather than relying on reactive enforcement.

- **Identification of Repeat Offenders and Accident-Prone Hotspots**
 - The system will help **pinpoint drivers or locations** that frequently contribute to traffic violations or accidents, enabling targeted interventions.
- **Cost-Effective Resource Allocation**
 - Predictive insights allow traffic authorities to **deploy enforcement personnel and equipment efficiently**, minimizing unnecessary resource expenditure.
- **Enhanced Public Safety and Reduced Accident Rates**
 - By identifying potential speeding violations in advance and supporting **dynamic traffic control**, the system contributes to **safer highways and fewer accidents**.

4.3 Scope and Deliverables

Scope:

- The research focuses on **predictive analysis of speeding violations** using RFID and IoT data.
- Implementation is **limited to major highways in Sri Lanka**, taking into account local traffic patterns, vehicle types, and driver behavior.
- The study emphasizes **realistic deployment scenarios** and practical applicability for traffic authorities.

Deliverables:

1. **Preprocessed Violation Dataset**
 - Cleaned, normalized, and feature-engineered data ready for machine learning analysis.
2. **Machine Learning Predictive Model**
 - A model capable of forecasting high-risk speeding events and categorizing them by driver, vehicle type, location, and time.
3. **Vehicle Classification System**
 - A module for classifying vehicle types using RFID and optionally image-based data to enhance prediction accuracy.
4. **Visualization and Reporting Dashboard**
 - Interactive dashboard for authorities to monitor trends, hotspots, and real-time predictions.
5. **Research Documentation and Evaluation**
 - Comprehensive documentation detailing **methodology, experimental results, and system evaluation**.

4.4 Research Constraints

The project also faces several **limitations and challenges**, which must be considered during development:

- **Limited Availability of Real-World RFID Datasets**
 - Actual RFID-based traffic violation data in Sri Lanka may be scarce, requiring reliance on simulated or secondary datasets.
- **High Variability in Traffic and Driver Behavior**
 - Mixed traffic conditions, including three-wheelers, buses, motorcycles, and varying driver behaviors, introduce **complexity in predictive modeling**.
- **Computational Demands for Training Machine Learning Models**
 - Large-scale datasets and complex models (e.g., ensemble methods or neural networks) require **high computational resources**, which may affect training and experimentation.
- **Legal and Privacy Concerns**
 - Handling driver identification data must comply with **privacy laws and regulations**, limiting certain data collection or usage options.
- **Environmental and Operational Factors**
 - Changing weather, lighting conditions, and road infrastructure variability may impact the accuracy and consistency of prediction models.

5.BUDGET AND BUDGET JUSTIFICATION

This research has been calculated to fit on its important parts such as hardware, software, data collection and validation parts included in a carefully estimated budget. The allocation will provide cost-effectiveness in terms of achieving the technical needs of successful implementation and testing under real-world conditions in the Sri Lankan traffic conditions.

Item	Cost Estimate (LKR)	Justification
Cloud Computing Resources (AWS/Collab Pro)	12000	For model training and deployment
RFID Chip and scanner	75000	Base 64 or MIFARE Series
Miscellaneous	10000	Internet and contingency expenses
Data Annotation Tools	15000	For labeling vehicle datasets
Chip Card Create for ID	10000	Each card had chip

Total Estimated Budget: LKR 132,000

Figure 2 Budget plan

Gantt chart

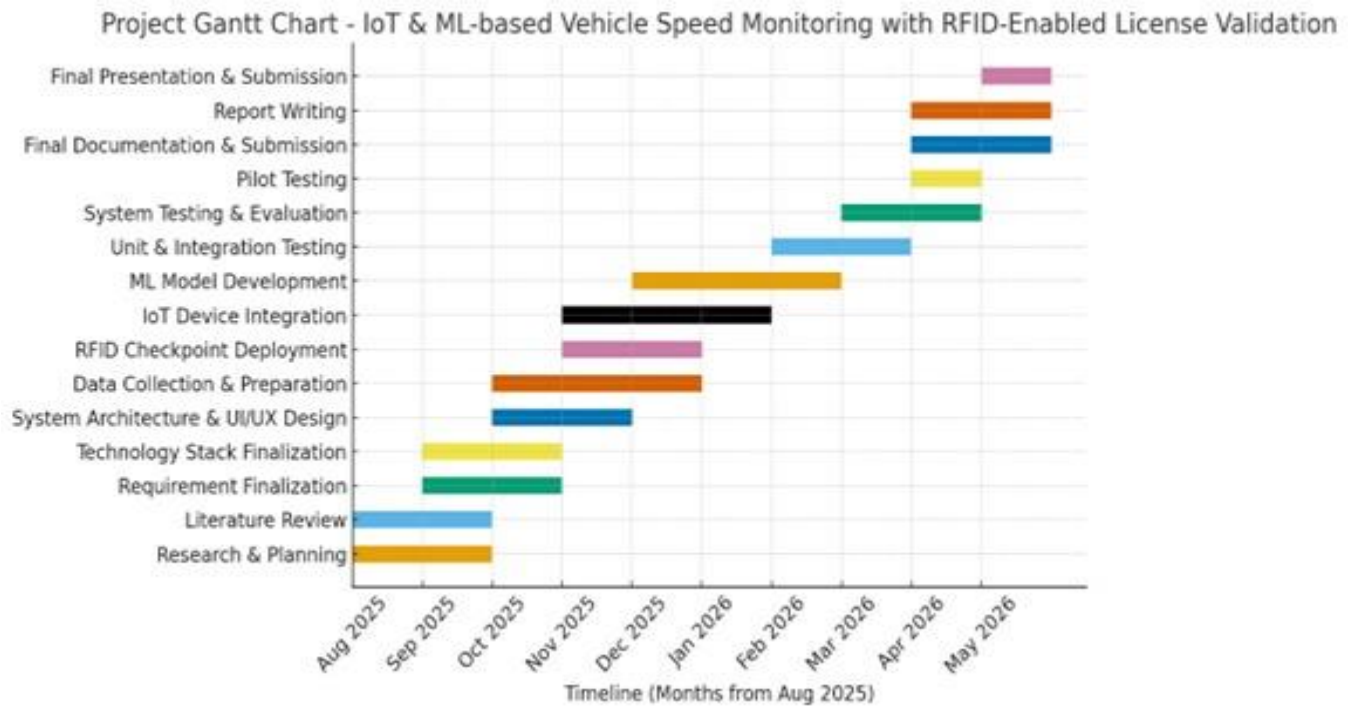


Figure 3 Gantt chart

WBS

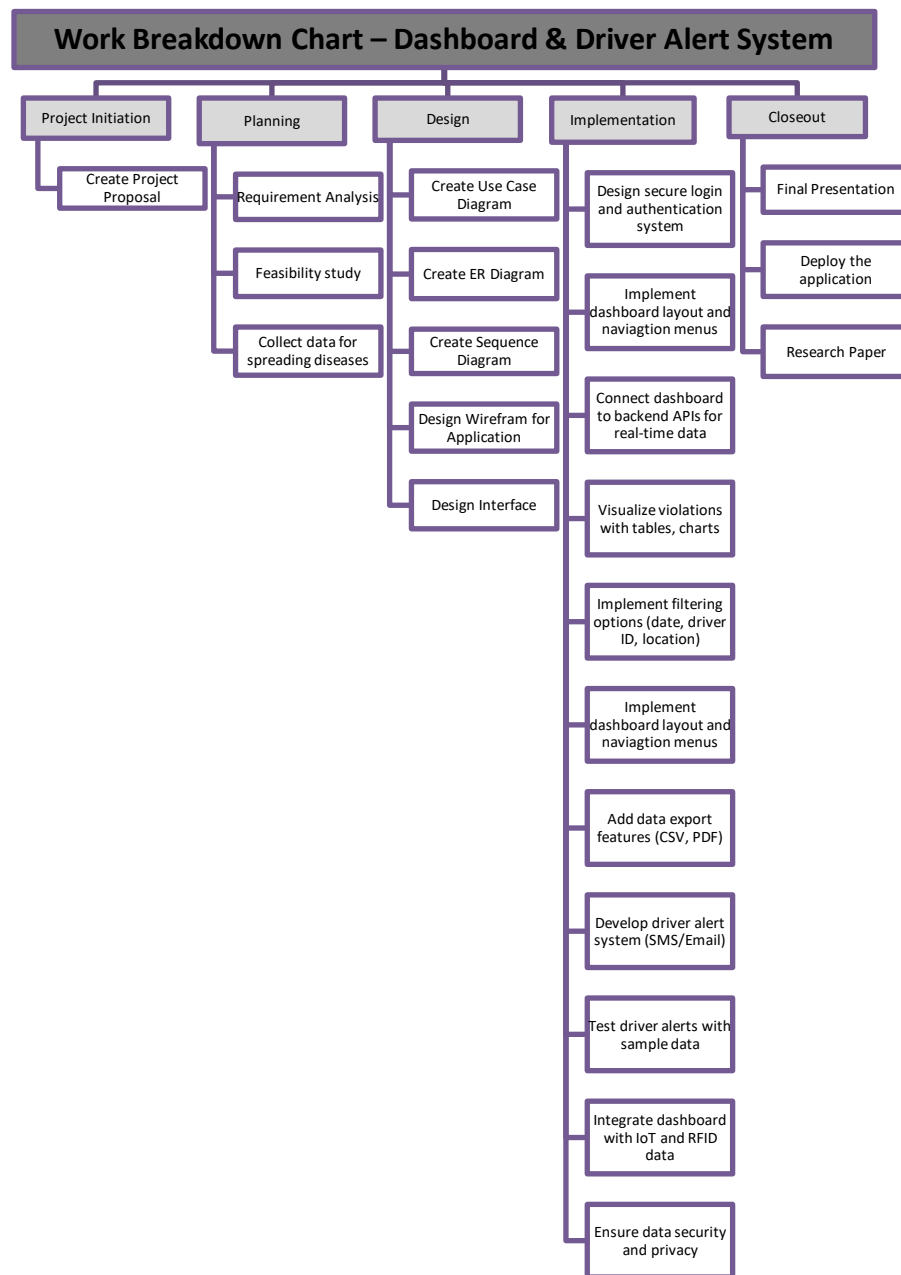


Figure 4 WBS diagram

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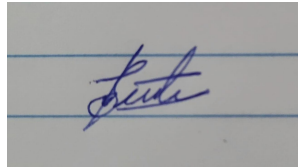
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	Title	Name	Signature
Supervisor	Prof.	Kapila Dissanayake	

Abstraction

Over-speeding is a leading cause of road traffic accidents and fatalities globally. According to the World Health Organization, approximately 1.3 million people lose their lives in road crashes each year [1], with speeding identified as a major contributory factor especially in low- and middle income countries [2]. Sri Lanka faces a severe road safety challenge due to rapid urbanization, increasing vehicle ownership, and limited automated enforcement measures. Existing methods like speed guns and stationary cameras are largely reactive, cover only specific spots, and lack predictive insight.

This research aims to develop the machine learning backend of an integrated RFID and IoT based smart speed monitoring system. In the proposed system, RFID readers on highways will detect speed violations by reading data from RFID tags (embedded in vehicles or driver licenses) and calculating vehicle speeds. The collected violation data will be processed using machine learning algorithms to discover patterns and 1 predict future high risk scenarios. Specifically, the model will analyze historical violation records to forecast factors such as high risk vehicle categories, time periods, and road segments prone to speeding incidents.

In addition to real time detection of speed limit violations, the system will provide a predictive analytics dashboard. This dashboard will present authorities with interactive charts and risk reports, enabling them to allocate enforcement resources more strategically and initiate targeted road safety campaigns. An image processing module for vehicle type classification is also considered to improve the accuracy of risk profiling. The novelty of this work lies in bridging the gap between enforcement and prediction: extending traditional speed monitoring with a data driven forecasting and alerting capability. The outcome is expected to transform traffic law enforcement in Sri Lanka from a reactive practice into a proactive, evidence based approach that can save lives and enhance road safety on highway.

Table of Contents

Declaration	2
Abstract	3
List of Figures	5
1.Introduction	6
1.1 Background & Literature Survey	7
1.2 Research Gap	8
1.3Research Problem.....	8
2.Objectives	10
2.1 Main Objective	10
2.2 Specific Objectives	10
3.Methodology	11
3.1 Project Overview	11
3.2 System Diagram	11
3.3Individual Component	12
4.Research & Development Overview	13
4.1 Data Sources	13
4.2 Anticipated Benefits	14
4.3 Scope and Deliverables	15
4.4 Research Constraints	16
5.Budget and Budget Justification	17
6.References	19

List of Figure

Figure 1 System diagram.....	12
Figure 2 Budget plan.....	19
Figure 3 Gantt chart	20
Figure 4 WBS diagram.....	21

1. INTRODUCTION

Rapid urbanization and motorization have led to worsening road safety in Sri Lanka. Speeding has been identified as a pivotal factor in fatal crashes globally, contributing to roughly one-third of road deaths in high-income countries and nearly half in developing countries [1]. In Sri Lanka, road accident fatalities have risen with the growth of vehicle usage; over 2,500 lives were lost in 2023 alone, making road injuries the second leading cause of death nationally [2]. These statistics underscore the urgent need for improved speed management and enforcement strategies.

Current enforcement mechanisms on Sri Lankan highways rely on manual or point-based systems (e.g, handheld speed guns and fixed speed cameras). While these tools serve as deterrents, they suffer from several limitations. They operate reactively penalizing speeding after it occurs and cover limited road sections. They are also labor-intensive and prone to human error, resulting in many speeding incidents going undetected. The lack of continuous monitoring and predictive insight means authorities cannot proactively identify when or where high-risk violations are likely to happen next [3].

Internationally, advancements in IoT and intelligent transportation systems have demonstrated the potential for more proactive traffic law enforcement. For instance, many countries have implemented RFID-based systems for toll collection and congestion charging (e.g., India's Fast tags and Singapore's Electronic Road Pricing), which demonstrates an infrastructure that could be extended to speed monitoring. However, the application of such predictive enforcement for speeding remains underexplored. In Sri Lanka's context, the use of RFID technology in traffic has so far been confined to applications like automated tolling, without integration into a predictive enforcement framework.

Machine learning offers powerful techniques to analyze traffic violation data for patterns and risk factors. Prior studies have applied algorithms such as Random Forests, Gradient Boosting, and Neural Networks to related problems like predicting accident hotspots or driver risk profiles with promising results. For example, Random Forest models have been shown to outperform other classifiers in predicting traffic rule violations based on driver behavior data [6]. Yet, very few implementations target the unique traffic conditions in Sri Lanka, which involve heterogeneous vehicle types (cars, buses, motorcycles, threewheelers) and often chaotic driving behaviors. This creates an opportunity to develop a tailored solution that learns from local violation patterns.

In summary, there is a clear gap in Sri Lanka's traffic enforcement system: the absence of a technology driven, predictive tool to complement existing speed monitoring efforts. By leveraging RFID-based identification of vehicles and machine learning analytics, the proposed system seeks to fill this gap. It will enable authorities not only to detect and penalize speed limit violations in real time, but also to foresee when and where such violations are most likely to occur. This proactive capability can inform more effective deployment of police patrols, smarter placement of mobile speed traps, and data-driven public awareness campaigns focused on speeding. Ultimately, the system aims to reduce accidents and save lives by turning reactive enforcement into preventive intervention.

1.1. Background & Literature Survey

Speed limit enforcement has been a subject of extensive research due to its impact on road safety. Globally, speeding is recognized as one of the leading contributors to road crashes [3]. The World Health Organization's reports have consistently highlighted that controlling vehicle speed is critical to preventing accidents and mitigating injury severity [1]. Traditional enforcement methods (such as traffic police surveillance, radar guns, and fixed cameras) have limitations in coverage and efficiency. These methods typically address violations at specific points and do little to predict or prevent future incidents.

Recent technological interventions in traffic management provide insight into how enforcement could be improved. IoT-based solutions have been used in various domains: for example, RFID technology underpins electronic toll collection systems like FASTag in India and smart road pricing in Singapore. In these systems, vehicles carry RFID tags that are read by sensors to automate transactions or monitor vehicle entry/exit. Extending this concept, researchers have explored using RFID for speed monitoring; by installing multiple RFID readers along a road, a vehicle's average speed between checkpoints can be calculated. If the computed speed exceeds the legal limit, a violation can be recorded. This approach can catch speeders over a stretch of road rather than a single point, addressing the issue of drivers slowing down only near known camera spots. However, purely RFID-based speed enforcement has challenges, including the need for widespread tag adoption and infrastructure deployment [4, 5].

Another strand of related work is the development of automated traffic violation detection systems using computer vision and other sensors. Advanced surveillance cameras and radar units can automatically detect speeding, red-light running, or other violations and instantly alert authorities or trigger automated responses. For instance, Asoba et al. (2020) proposed a system combining IoT devices and image processing algorithms to improve the accuracy of detecting violations like speeding or illegal crossings, thereby automating the penalty process [7]. These approaches demonstrate the feasibility of an end-to-end automated enforcement pipeline.

Machine learning is increasingly at the core of predictive traffic safety applications. Studies focusing on crash and violation prediction have utilized data-driven models to identify patterns that precede incidents. Driver profiling using historical data is one approach: Jose et al. (2023) developed a system that predicts the likelihood of a driver committing violations based on personal and behavioral factors, where Random Forest yielded the highest prediction accuracy [6]. Others have applied neural network models to time-series traffic data to forecast accident hotspots or peak violation periods [8]. These efforts show that with sufficient data, it is possible to forecast risk levels with useful accuracy, enabling preemptive actions by authorities.

Despite these advancements, there remains a lack of systems that combine RFID-based monitoring with machine learning for proactive speed violation enforcement, especially tailored to Sri Lankan roads. This research therefore builds upon the above strands RFID IoT infrastructure for data collection, AI-based violation detection, and predictive analytics to create a unified solution for smart speed enforcement.

1.2. Research Gap

While modern traffic monitoring technologies are being adopted in many countries, Sri Lanka still lacks a predictive enforcement tool for speed violations. Current applications of RFID in Sri Lanka are mostly limited to automated toll collection and basic vehicle identification tasks. Real-time IoT-based traffic monitoring initiatives (such as smart traffic lights or CCTV networks) exist in some urban areas, but these generate data that is not yet leveraged for prediction. In enforcement, authorities predominantly use reactive measures penalizing drivers after a violation has occurred, without analysis of when the next violation might happen or which drivers are likely to re-offend.

The gap identified is the absence of an integrated platform that transforms raw violation data into actionable intelligence. With the proliferation of IoT sensors, there is an opportunity to gather rich datasets on vehicle speeds and behavior. However, without machine learning analytics, this data remains underutilized for strategic decision-making. For example, a predictive model could warn if a particular highway segment is likely to experience a surge in speeding incidents during certain hours, or if certain drivers (such as repeat violators) pose a high risk of future violations. Such capabilities are not part of the current Sri Lankan traffic enforcement regime.

In summary, the research gap lies in bridging the current enforcement prediction divide. There is a need for a system that not only detects speeding violations in real time but also learns from accumulated data to forecast future violations. By addressing this gap, enforcement agencies can move from a solely punitive approach to a preventative one, improving road safety outcomes.

1.3. Research Problem

Research Question: How can data from an RFID based speed monitoring system be effectively utilized with machine learning to predict future high-risk speeding patterns categorized by driver identity, vehicle type, location, and time so that traffic authorities in Sri Lanka can shift from reactive enforcement to a proactive strategy?

- **Data Acquisition and Integration:** Modern highways outfitted with RFID readers can capture vehicle information in real time. Each passing vehicle with an RFID tag (e.g., embedded in the driver's license or mounted on the vehicle) generates a record with attributes such as a unique vehicle/driver ID, vehicle type, timestamp, and speed (derived from travel time between reader checkpoints). The challenge is to gather a large, reliable dataset from these IoT devices for analysis.
- **Data Preprocessing:** Raw RFID data is often noisy, incomplete, or inconsistent. This project will focus on cleaning and normalizing the collected data so that it can be effectively used by machine learning models. Key steps include handling missing or erroneous readings, synchronizing timestamps, and engineering features such as violation frequency per driver, average speed per road segment, time-of-day indicators, and traffic density estimates. Properly preprocessed data ensures the model can detect meaningful patterns rather than artifacts of poor data quality.

- **Predictive Modeling:** The core challenge is to build accurate machine learning models that can identify patterns in historical speeding data and predict future violations. We will experiment with algorithms like Random Forest (for its robustness and interpretability), Gradient Boosted Trees (for capturing nonlinear relationships), and Neural Networks (for their ability to model complex interactions). The model will be trained to forecast high-risk scenarios, for example: which vehicle types or drivers are likely to overspeed, during what time windows, and on which highway segments. Performance metrics such as accuracy, precision/recall, and F1-score will be used to evaluate the models, and the best-performing model will be selected for deployment.
- **Risk Profiling:** Beyond individual predictions, the system will categorize and profile risk factors. It will generate insights like “Driver X has a high likelihood of repeated violations,” or “Highway segment Y is prone to speeding on weekend nights.” By grouping predictions into categories (driver, vehicle type, location, time), the system can produce a risk matrix that highlights where proactive measures are most needed. For instance, heavy vehicles might be found to violate speed limits mostly during off-peak hours, or certain locations might emerge as early morning hotspots for speeding.
- **Proactive Enforcement Mechanism:** The ultimate aim is to enable traffic authorities to take preventive action using these predictions. If the system predicts a surge in violations at a particular location and time, enforcement agencies could deploy mobile units or set up temporary warning signs in advance. Similarly, if certain drivers are flagged as high risk (e.g., those with multiple past violations in a short period), authorities could send warning notices or require those drivers to attend safety workshops. This shifts the paradigm from penalizing offenses after they occur to actively working to prevent them.

Addressing the above facets of the problem will demonstrate how an RFID + ML based approach can transform raw traffic data into actionable enforcement strategies. Success will be measured by the system’s ability to accurately predict high-risk situations and the potential impact of these predictions on improving road safety.

2. OBJECTIVES

The overarching goal of this project is to design and implement a smart highway speed monitoring and predictive enforcement system that integrates RFID-based vehicle identification, IoT-enabled speed sensing, and machine learning analytics. By achieving this, we aim to enhance real-time violation detection and enable authorities to anticipate and prevent future speeding incidents.

2.1 Main Objective

Develop a comprehensive system combining RFID technology, IoT infrastructure, and machine learning to automatically detect speed limit violations on highways, alert law enforcement in real time, and generate predictive insights (such as identifying high-risk drivers, locations, and time periods) for proactive traffic safety management.

2.2 Specific Objectives

- **Predictive Modeling:** Create a machine learning model capable of analyzing historical RFID captured speeding data to predict future violation trends and identify patterns (e.g., frequent offenders, peak violation hours, accident-prone road segments).
- **Vehicle Classification Enhancement:** Incorporate an image processing component (using techniques like Convolutional Neural Networks) to accurately classify vehicles by type from camera or video data, in order to enrich the violation dataset and improve the model's context (e.g., distinguishing between car, bus, truck behaviors).
- **Real-Time Notification System:** Design the backend service to receive speed readings from RFID checkpoints and immediately determine if a violation has occurred. Implement logic to compare calculated speeds against legal speed limits for each road zone, and trigger instant alerts via SMS/ email (using services like Twilio) to notify nearby law enforcement units of violations.
- **Data Storage and Management:** Deploy a secure database (e.g., Firebase or MongoDB) to store all violation records and related data. Ensure this database can handle large volumes of data and allows efficient querying for analytics and reporting.
- **Visualization Dashboard:** Develop an interactive dashboard for traffic authorities that displays realtime violation alerts on a map interface and provides analytical charts of historical data. This dashboard should highlight trends such as the most common violation times, locations with frequent speeding, and profiles of repeat violators, thereby assisting in strategic planning.
- **System Validation:** Test the integrated system with a combination of simulated data (and any available real-world data) to validate performance. This includes verifying the accuracy of speed calculation from RFID data, the precision of the machine learning predictions (comparing predicted high-risk instances against actual data), and the responsiveness of the alert dispatch mechanism. Based on validation results, fine-tune the system and provide recommendations for deployment by enforcement agencies.

3. METHODOLOGY

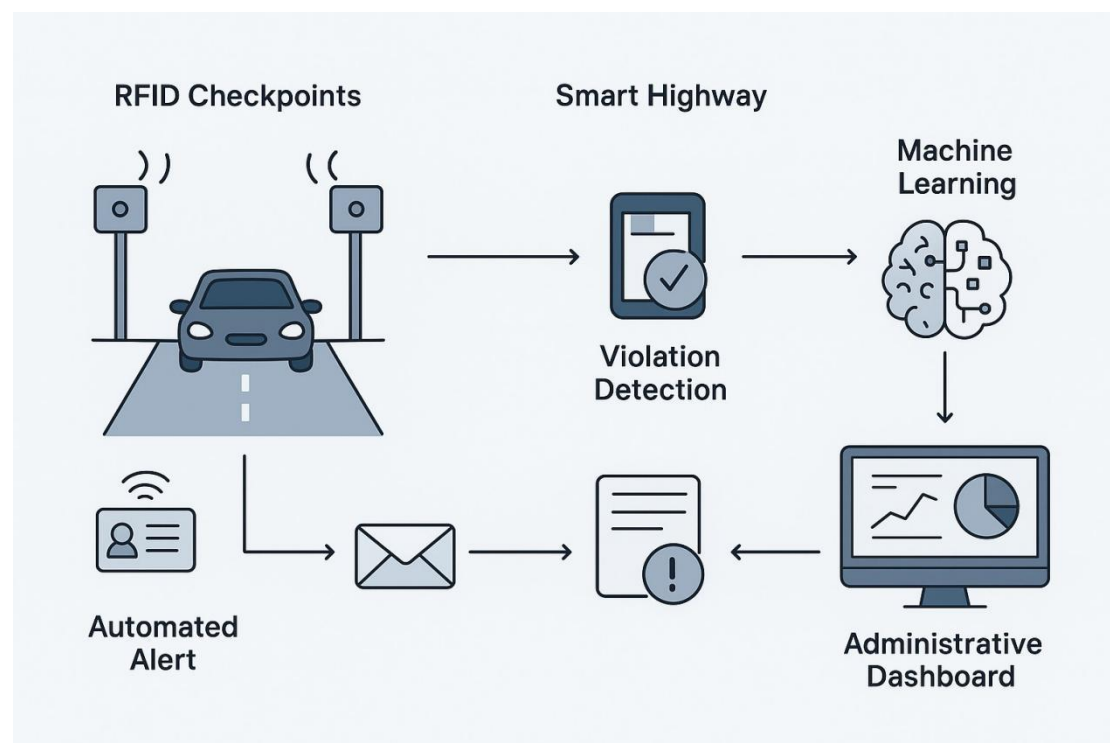
3.1 Project Overview

The methodology of this project follows a data-driven development pipeline, encompassing hardware deployment, data processing, machine learning modeling, and user interface design. First, a network of RFID readers will be conceptually placed at strategic points on a highway (for example, at the beginning and end of a highway segment or near toll booths). Each vehicle carries an RFID tag that is read at these checkpoints, enabling the system to calculate its average speed over the known distance between readers. These speed readings, along with the vehicle's ID, timestamp, and location, form the raw dataset of potential violations (a violation occurs when the average speed exceeds the legal threshold).

The collected data will be transmitted via an IoT gateway to a central server or cloud platform. There, data preprocessing routines will clean and organize the information. The machine learning module will then analyze this historical data to learn patterns and make predictions about future speeding occurrences. Finally, a web-based dashboard will present both real-time and predictive information to end-users, and a notification service will handle instant alerts to enforcement officers. Figure 1 illustrates the high-level architecture of the proposed system, from data capture on the highway to back-end analysis and front end output.

3.2 System Diagram

Figure 1 illustrates the system architecture for the proposed model.



3.3 Individual Components

3.3.1 IoT Speed Sensing Unit

The IoT speed sensing unit is the physical layer of the proposed system. Two UHF RFID readers will be installed at fixed checkpoints along a highway segment, each paired with a panel antenna for reliable tag detection. These readers are connected to an ESP32 microcontroller through UART, which timestamps each tag reading using an NTP-synchronized clock. When a vehicle passes through both checkpoints, the ESP32 calculates its speed by dividing the fixed distance between the checkpoints by the time difference (Δt) between tag detections. To ensure real-time operation, the computed speed and tag ID are formatted into a JSON packet and transmitted to the backend service via HTTP over Wi-Fi. This component provides a low-cost, tamper-resistant alternative to radar or camera-based speed monitoring, ensuring accurate driver identification through RFID-embedded licenses.

3.3.2 Backend Service

The backend service forms the central processing and decision-making component of the system. It is designed as a RESTful API that receives speed and tag data from each IoT checkpoint. The backend applies speed violation logic specific to each road segment, ensuring that different zones with unique speed limits are enforced accurately. Once a violation is detected, the record is securely stored in a cloud-hosted NoSQL database such as Firebase Firestore or MongoDB Atlas. In addition to storage, the backend is also responsible for real-time notifications. Using services like Twilio for SMS and SendGrid for email, alerts containing the vehicle ID, speed, location, and time of violation are immediately dispatched to relevant traffic authorities. By automating this workflow, the backend ensures enforcement without human intervention and provides a scalable architecture for future expansion across multiple highways.

3.3.3 Machine Learning Module

The machine learning module focuses on predictive analysis, turning historical violation records into actionable insights. Violation datasets are preprocessed to remove noise, normalize speed and time data, and extract features such as driver frequency, vehicle type, time of day, and location. Multiple classification algorithms—including Random Forest, Gradient Boosting, and Neural Networks—are trained to forecast potential speeding events. Additionally, clustering techniques such as K-Means are used to identify patterns of driver behavior, including repeat offenders or high-risk vehicle groups. The trained models generate predictive outputs, such as identifying peak violation hours or high-risk highway segments, which are then exposed to the dashboard through APIs. This component extends the system beyond real-time detection into proactive enforcement planning.

3.3.4 Visualization Dashboard

The visualization dashboard provides the user interface for traffic authorities to monitor and analyze violations. Designed as a secure web application, the dashboard displays real-time updates from the backend in the form of maps, heatmaps, and statistical charts. Authorities can filter records by time, location, or vehicle ID, and generate downloadable reports in formats such as CSV or PDF for recordkeeping. Importantly, the dashboard also integrates predictive insights from the machine learning module, highlighting risk-prone areas and time periods. By combining live alerts with long-term predictive analysis, the dashboard empowers decision-makers to allocate enforcement resources strategically and launch targeted awareness campaigns.

4. RESEARCH & DEVELOPMENT OVERVIEW

4.1 Sources of Data

Developing the proposed system requires relevant and sufficient data for both training the machine learning model and evaluating system performance. Several data sources will be utilized:

- **Simulated RFID Datasets:** Since real-world RFID speed violation data from Sri Lankan highways may not be readily available, synthetic datasets will be generated to simulate vehicle movements, speeds, and violation patterns. This simulation allows testing the system under controlled conditions and can cover a wide range of scenarios (e.g., varying traffic volumes, different times of day, special events) that might be challenging to capture in a limited real dataset.
- **Secondary Traffic Data Repositories:** Publicly available traffic datasets and prior studies' data will be considered for model benchmarking. For instance, we might use data from international projects or literature where IoT devices were used to monitor traffic speeds or where speeding violations were recorded (such as datasets from smart city pilot programs). These external datasets help in validating the model's performance and ensuring that it aligns with known patterns (for example, verifying that predicted peak speeding times coincide with typical rush hours observed elsewhere).
- **Annotated Image Datasets:** If the image processing module for vehicle classification is included, we will use existing image datasets of vehicles to train and test the CNN. Examples include the UADETRAC dataset or COCO dataset (which have labeled vehicles), or even custom images captured from local roads if accessible. These images will be annotated by vehicle type to serve as ground truth for the classification algorithm.
- **WHO and Government Statistics:** Baseline statistics on road traffic accidents, fatalities, and the contribution of speeding will be gathered from the World Health Organization and Sri Lankan government reports. These statistics provide context and justification for the project (e.g., underscoring why speeding is a problem worth addressing) and also help in setting targets (for example, if speeding contributes to 40% of accidents, the system's long-term goal might be to reduce that figure significantly). Additionally, any available data on current enforcement outcomes (like number of speeding tickets issued per month in certain areas) can be used to compare against the system's detection rate.

4.2 Anticipated Benefits

Implementing the smart speed monitoring system is expected to yield multiple benefits.

- **Proactive Traffic Enforcement:** By predicting where and when speeding violations are likely to occur, authorities can shift to a proactive stance. Instead of waiting for an accident or violation to happen, they can position resources (patrol units or mobile speed cameras) in anticipation of high-risk conditions. Over time, this could lead to deterrence – if drivers know that enforcement might be around the next corner at any time, not just at fixed camera spots, they may be more inclined to obey speed limits consistently.
- **Identification of Repeat Offenders and Hotspots:** The system's analytics will help pinpoint drivers who frequently violate speed limits and road segments that consistently see high rates of speeding. This information is valuable for targeted interventions. For example, repeat offenders might be required to undergo additional driver training or face escalating penalties. High-risk locations might be candidates for engineering solutions (better signage, speed bumps) or increased monitoring during specific hours. Recognizing these patterns means enforcement and safety measures can be tailored rather than one-size-fits-all.
- **Efficient Resource Allocation:** Traffic enforcement agencies often have limited manpower and equipment. The insights from the predictive model and dashboard can guide decision-makers in deploying these resources more efficiently. For instance, if the system shows that early Monday mornings have very low violation rates on a certain highway, enforcement can be relaxed at that time in favor of other tasks, while a spike on Friday nights would signal a need for more officers or sobriety checkpoints. This data-driven allocation can improve the overall effectiveness of the traffic police force without needing additional budget.
- **Enhanced Public Safety and Reduced Accidents:** The ultimate measure of success will be a reduction in speed-related accidents and fatalities. By catching violations in real time and deterring would-be violators through increased enforcement presence (informed by predictions), the system aims to reduce the incidence of extreme speeding. Even a modest decrease in average traffic speeds can significantly lower crash severity. Over months and years, this could translate into lives saved and injuries prevented. Moreover, the presence of a “smart” enforcement system has a psychological effect; public awareness campaigns about the system (e.g., “highways are monitored by smart sensors”) can encourage more compliant driving behavior.
- **Data-Driven Policy and Planning:** Beyond enforcement, the data gathered can inform broader transportation policy. For example, if data shows that certain times or locations have habitual speeding because of lower traffic (and perhaps unrealistically low posted speed limits), authorities might consider adjusting speed limits or improving road design. The evidence base provided by the system can support proposals for road infrastructure improvements, funding requests for more IoT installations, or changes in legal frameworks (such as demerit point systems for repeated speeding).

4.3 Scope and Deliverables

Scope: The scope of this project is focused on the development and demonstration of the technological solution, rather than full nationwide deployment. It will concentrate on highways (as opposed to urban city streets, which have different dynamics like traffic lights and intersections). The project will primarily involve software development (simulation environment, database, machine learning, dashboard) and possibly a small-scale hardware demo (using a few RFID readers/tags in a controlled environment or lab). Considerations specific to Sri Lanka – such as common vehicle types, typical driver behaviors, and local road conditions – will be incorporated into model assumptions and simulation parameters to ensure the solution is realistic.

Deliverables: By the end of the project, the following deliverables are expected:

1. **RFID Speed Monitoring Prototype:** A proof-of-concept setup demonstrating how RFID readers and tags can be used to capture vehicle speed. This might be illustrated through a video or live demo where two RFID readers simulate a checkpoint, and a tagged object's speed is calculated.
2. **Processed Violation Dataset:** A compiled dataset (in CSV or similar format) containing timestamped entries of vehicle passes, calculated speeds, and flag indicating whether a violation occurred. This will include both simulated data and any real sample data if obtainable.
3. **Trained Machine Learning Model:** The final machine learning model (code and saved model file) along with documentation of its performance (e.g., confusion matrix, accuracy metrics on validation data). This will demonstrate the model's ability to predict high-risk scenarios and will be accompanied by example use-cases (such as feeding in a certain day's data to predict the next day's high-risk periods).
4. **Vehicle Classification Module:** A separate component that can take an image and output a vehicle type classification. Deliverable includes the trained CNN model and a brief report on its accuracy.
5. **Backend Service and APIs:** A running backend server (or a simulation of it) that ingests RFID readings, runs the violation detection logic, interfaces with the ML model, and triggers alerts. API documentation will be provided, describing how data flows in and out (for instance, an endpoint that can be called by an RFID reader to send a read, and an endpoint that the dashboard calls to get prediction data).
6. **Alerting System Demonstration:** A demonstration (possibly simulated) of the real-time alert dispatch. For example, using a service like Twilio in a test environment to show that when a violation is detected, an SMS is sent to a specified phone number with the violation details.
7. **Visualization Dashboard:** A functional dashboard front-end that authorities can use. This will be delivered as a web application link or screenshots if a live deployment is not feasible. The interface will be intuitive and include the features described (maps, charts, filters, etc.). It will be populated with sample data to illustrate its capabilities.

8. **Project Report and Technical Documentation:** A comprehensive report (including this proposal and subsequent final thesis) documenting the research process, system design, implementation details, results, and recommendations. Additionally, a user manual or technical guide for the system will be provided, explaining how to deploy and use each component.

4.4 Research Constraints

Despite careful planning, the project will have to navigate several constraints and challenges.

- **Limited Availability of Real Data:** Access to actual highway speed violation data or RFID records in Sri Lanka may be very limited due to privacy or operational restrictions. This means the project might rely heavily on simulated data. While simulation can approximate reality, it may not capture unpredictable human behaviors or rare events. This constraint will be mitigated by calibrating simulations with any known statistics (like average traffic speeds from literature) and clearly stating assumptions in the analysis.
- **Technical Infrastructure Constraints:** Deploying RFID readers on highways and equipping vehicles with RFID tags is a large-scale effort beyond this project's scope. As such, the hardware demonstration will be limited. The success of the project conceptually assumes such infrastructure can be in place. Any hardware prototypes (e.g., using short-range RFID in a demo) might face issues like limited range or interference, which will need to be acknowledged.
- **Computational Requirements:** Training the machine learning model, especially if using neural networks or processing image data, could be computationally intensive. There might be constraints on processing power available (the project budget includes some cloud computing credits, but they are finite). If the dataset is very large or the model is complex, training times could be long, affecting iterative improvement. We will address this by optimizing code, using efficient algorithms, and if needed, simplifying the model to fit resource constraints.
- **Privacy and Legal Considerations:** The use of RFID to monitor vehicles touches on privacy issues. Even though this system is for law enforcement, data like travel patterns of individual vehicles are sensitive. The project will ensure that any personal data is anonymized in analysis (e.g., using hash IDs instead of actual license numbers) and discuss the need for data protection measures if deployed (such as secure data storage and strict access controls). Legally, automated enforcement (e.g., issuing tickets via an automated system) may require changes in traffic laws. While that is beyond the project's implementation, the proposal and final recommendations will consider what legal framework would be needed to allow the system's outputs (predictions or detected violations) to be used in an official capacity.
- **Environmental and External Factors:** The system's efficacy could be influenced by factors outside the direct control of technology. For instance, extreme weather (heavy rain, fog) could affect RFID read reliability or camera visibility. Drivers might behave differently during festive seasons or fuel shortages, which a model might not predict if such scenarios weren't in the training data. Additionally, if drivers become aware of the system's presence, they might alter their behavior (which is good for safety but can change the data patterns). The project will note these uncertainties and design the system to be as adaptable as possible (for example, including weather data as an input to the model if available, so it learns that during rain, average speeds drop).

5. BUDGET AND BUDGET JUSTIFICATION

This project's budget has been carefully estimated to cover all crucial components, including hardware, software, data handling, and validation activities. The allocation is designed to be cost-effective while ensuring the technical needs for successful implementation and testing in Sri Lankan traffic conditions are met.

Item	Cost Estimate (LKR)	Justification
Cloud Computing Resources (AWS/Collab Pro)	12000	For model training and deployment
RFID Chip and scanner	75000	Base 64 or MIFARE Series
Miscellaneous	10000	Internet and contingency expenses
Data Annotation Tools	15000	For labeling vehicle datasets
Chip Card Create for ID	10000	Each card had chip

Total Estimated Budget: LKR 132,000

Figure 2 Budget plan

Gantt chart

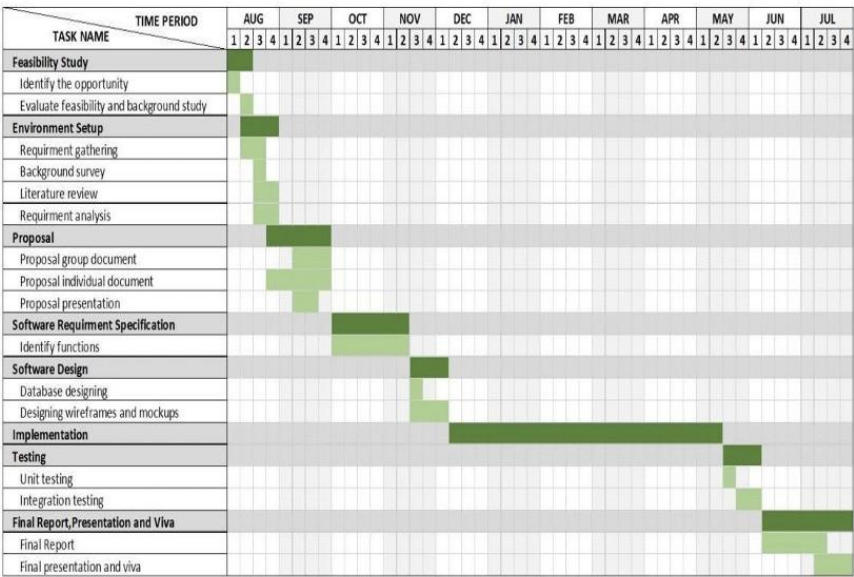


Figure 3 Gantt chart

WBS

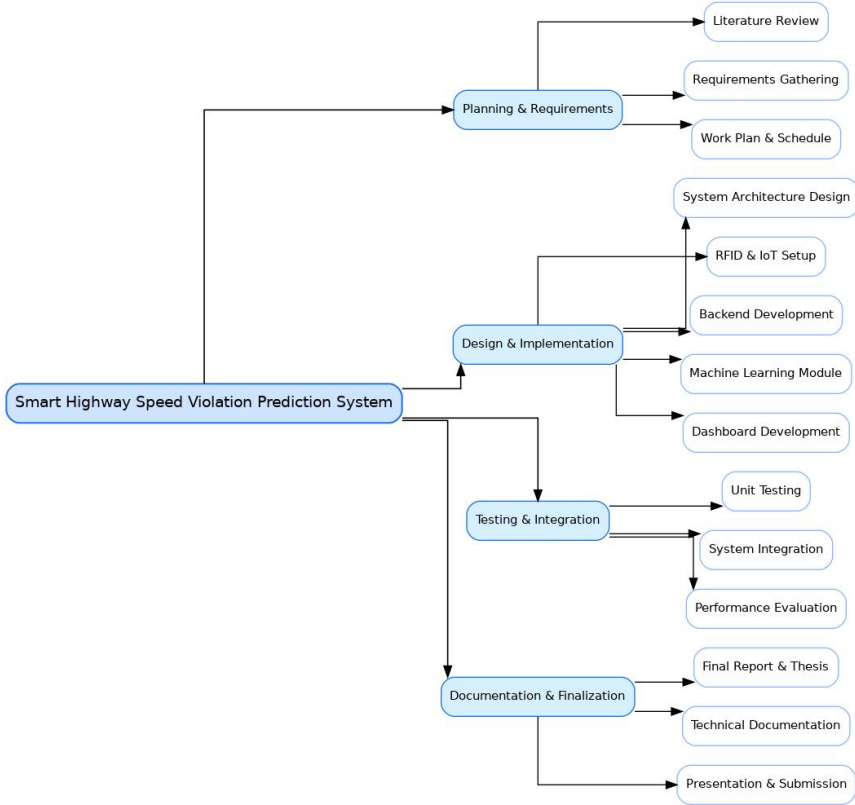


Figure 4 WBS diagram

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IOT & ML-BASED VEHICLE SPEED MONITORING WITH RFID-ENABLED LICENSE VALIDATION

Project Proposal Report

Project ID: 25-26J-314

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IT22308252

BSc (Hons) Degree in Information Technology Specialized in
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August 2025

Declaration

To the best of my knowledge and belief, this proposal does not contain any previously published or written by another person material, except where the acknowledgement is made in the text. I declare that this is solely my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning.

Group Member name	Student ID	Signature
PERERA A.K.I.R	IT22308252	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Name of supervisor: Dr. Kapila Dissanayake

Name of co-supervisor: Ms. Suranjanee De Silva

	Title	Name	Signature
Supervisor	Dr.	Kapila Dissanayake	

Abstraction

Road accidents caused by overspending continue to be one of the leading causes of road fatalities in Sri Lanka. Although existing systems such as speed guns and cameras are used to monitor violations, they are limited in coverage, require heavy manual monitoring, and provide very little predictive insight. To overcome these gaps, the main project proposes an IoT and Machine Learning-based speed monitoring solution using RFID-enabled licenses. Within this system, my role focuses on the design and development of a Dashboard and Driver Alert System which will act as the central interface for enforcement and awareness.

The dashboard will be a secure web-based application that allows traffic authorities to log in and monitor overspending violations in real time. Data collected from RFID checkpoints and IoT devices will be displayed through tables, and graphs, helping authorities identify high-risk locations, time periods, and vehicle types. In addition, the dashboard will provide filtering options by date, road segment, or driver ID, along with export functions to generate reports in formats such as CSV and PDF. These features will support data-driven enforcement and make traffic planning more effective.

Another important part of my work is the Driver Alert System, which will directly notify drivers about their overspending through SMS or email. This feature is designed not only to inform drivers of their violations but also to encourage better driving behavior and reduce repeat offences. Together, the dashboard and alert mechanism create a bridge between enforcement authorities and drivers, enabling real-time action as well as long-term road safety improvements.

By combining real-time monitoring, predictive analysis, and direct driver engagement, this research aims to contribute a practical and scalable solution that enhances traffic enforcement, supports policymaking, and promotes safer highways in Sri Lanka.

Table of Contents

Declaration.....	2
Abstraction.....	3
List of Figure	5
1. INTRODUCTION	6
1.1. Background & Literature Survey.....	7
1.2. Research Gap	8
1.3. Research Problem	9
2. OBJECTIVES	10
2.1 Main Objective	10
2.2 Specific Objectives	10
3. METHODOLOGY	11
3.1 Project Overview	11
3.2. System diagram	12
3.2 Individual Component	13
4. RESEARCH & DEVELOPMENT OVERVIEW	15
5. BUDGET AND BUDGET JUSTIFICATION	17
References	20

List of Figure

Figure 1 System diagram.....	12
Figure 2 Budget plan	19
Figure 3 Gantt chart.....	20
Figure 4 WBS diagram	21

1. INTRODUCTION

Road traffic accidents are one of the leading causes of fatalities worldwide, with overspending being a major contributing factor. According to the World Health Organization, speeding accounts for more than 30% of road traffic deaths globally, particularly in low- and middle-income countries where enforcement remains a challenge [1]. In Sri Lanka, traditional methods such as handheld speed guns and speed cameras have been deployed, but these are limited in coverage, costly to maintain, and largely reactive [2]. Authorities also lack centralized platforms to monitor violations in real time or generate meaningful insights from recorded data.

The overall project addresses this issue by proposing an IoT and Machine Learning-based Vehicle Speed Monitoring System with RFID-enabled license validation. The system enables the automatic detection of overspending vehicles, data collection, and predictive analysis of traffic violation patterns. Within this broader project, my responsibility focuses on developing the Dashboard and Driver Alert System, which bridges enforcement data and real-time decision-making.

The dashboard will act as the central monitoring interface for traffic authorities, providing secure login access and organized displays of violation records. Unlike existing manual reporting methods, the dashboard will present real-time data using tables and statistical charts, enabling authorities to identify high-risk areas, times, and repeat offenders [3]. To support efficient enforcement, the system will include filtering options (e.g., by date, driver ID, or location) and allow officers to export violation records into formats such as CSV or PDF for official reporting.

Another key contribution is the Driver Alert System, which directly notifies drivers of their violations through SMS or email. Research shows that driver alert mechanisms increase awareness and encourage behavioral change, reducing repeat offences and improving compliance with traffic laws [4]. By integrating both authority-facing and driver-facing tools, this system aims to create a two-way solution: enforcement agencies gain real-time insights for policy and planning, while drivers are reminded of their responsibility to drive safely.

Therefore, the Dashboard and Driver Alert System strengthens the project's goal of reducing accidents caused by overspending by combining real-time monitoring, effective visualization, and driver engagement into a practical and scalable solution for Sri Lanka's highways.

1.1. Background & Literature Survey

Road traffic accidents caused by overspending continue to be a pressing issue both globally and in Sri Lanka. According to the World Health Organization, overspending contributes to over 30% of road traffic deaths worldwide, highlighting the need for stronger enforcement methods [5]. In Sri Lanka, conventional approaches such as handheld speed guns and static cameras have been implemented, but they remain limited in coverage, expensive, and heavily dependent on manual intervention [6]. These methods are also reactive, as they only detect violations after they occur, without providing predictive insights into recurring risks.

With the growth of smart technologies, IoT and RFID systems have been applied in traffic management. Countries such as India and Singapore already use RFID for toll collection and vehicle tracking, proving that the technology is scalable and reliable [7]. Research in IoT-based traffic monitoring shows that integrating real-time data with centralized systems helps reduce human dependency and allows authorities to act more effectively [8]. However, many existing systems focus mainly on detection, with less attention given to how the data is presented and used by traffic authorities for long-term planning.

The use of dashboards and visualization tools has been recognized as an effective way to make traffic data understandable and actionable. Dashboards provide quick access to real-time violations, visual trends, and summary reports, which help law enforcement agencies manage resources more effectively [9]. For example, Ullah et al. [10] developed a smart traffic management system where authorities could monitor traffic flow using a centralized interface, improving situational awareness. However, most implementations in Sri Lanka are still in early stages, with limited integration of predictive insights and reporting functions.

On the driver's side, alert systems such as SMS or in-vehicle notifications have shown positive effects in changing driving behavior. Studies indicate that timely feedback increases driver awareness and reduces repeat offences [11]. Caird et al. [12] found that real-time alerts can significantly reduce speeding tendencies, supporting the idea that direct communication with drivers can improve compliance with road safety laws.

Despite these developments, there is still a research gap in Sri Lanka's traffic enforcement context. Current systems lack: A centralized dashboard to display real-time overspending data in an organized and visual form, Filtering and exporting functions that help authorities analyze and report violations, Driver alert mechanisms directly linked with enforcement systems to improve awareness.

Therefore, this research proposes a Dashboard and Driver Alert System as part of the larger RFID-based monitoring project. This component bridges the gap between enforcement and awareness by giving authorities actionable insights through data visualization, while also engaging drivers with timely alerts to encourage safer driving.

1.2. Research Gap

Over speeding is one of the main reasons for road accidents in Sri Lanka. Existing enforcement tools such as speed guns and cameras are already in use, but they are limited in coverage and require manual monitoring, making them less effective in the long run [13]. These methods also fail to provide predictive insights that can help identify repeat offenders or high-risk road segments.

Although RFID technology has been successfully used in toll collection and vehicle identification in countries like India and Singapore [14], such systems mainly focus on identification and payments. They do not provide authorities with visual tools for real-time monitoring, nor do they include features such as filtering, reporting, or predictive analysis of speeding violations.

Another major difference is in driver awareness. Research shows that real-time feedback to drivers can improve compliance with traffic rules and reduce repeat offences [15]. However, in Sri Lanka, drivers are not informed immediately about their violations, which reduces the chances of changing unsafe driving behavior.

Therefore, the research gap lies in the lack of a centralized dashboard that allows traffic authorities to view and analyze violations in real time, combined with a driver alert system that provides immediate feedback. This project aims to fill that gap by developing a dashboard with visualization, reporting, and alert features that support both enforcement and driver awareness.

1.3. Research Problem

Over speeding has become one of the major causes of accidents and road fatalities in Sri Lanka. Even though some enforcement methods such as speed guns and static cameras are used, these solutions have several limitations. They only cover certain locations, require manual monitoring, and mainly provide reactive responses after the violation has already occurred. As a result, authorities cannot identify repeat offenders, predict high-risk times, or take preventive measures effectively. This limitation in existing enforcement reduces the overall impact of such systems.

Another problem is the lack of a centralized and user-friendly platform where traffic authorities can view, analyze, and manage data related to overspending violations. At present, most violation data is scattered or manually recorded, which makes it difficult to generate reports or recognize violation patterns. Without proper visualization tools such as charts, and statistical summaries, decision-making becomes slow and less effective. This creates a gap between raw data collection and meaningful enforcement actions.

On the drivers' side, there is also a clear weakness in the current system. Drivers are usually not informed about their violations until much later, often only when they are penalized. This delay reduces the opportunity for drivers to correct their behavior and avoid repeating the same offence. A real-time Driver Alert System that immediately notifies drivers of their speeding could encourage awareness and safer driving practices, but such a feature does not currently exist in Sri Lanka's traffic enforcement process.

Therefore, the main research problem is the absence of a centralized dashboard and driver alert mechanism that connects enforcement data with both traffic authorities and drivers in real time. Without these tools, enforcement agencies cannot fully utilize the collected data for proactive decision-making, and drivers are left unaware of their violations. Addressing this problem through a combined dashboard and alert system can improve both law enforcement efficiency and driver behavior, ultimately reducing overspending-related accidents.

2. OBJECTIVES

The main objective of this part of the study is to design and develop a Dashboard and Driver Alert System that provides traffic authorities with real-time access to overspending violations while also notifying drivers about their offences immediately. This component aims to bridge the gap between enforcement and driver awareness by combining a secure web dashboard with data visualization, filtering, and reporting features, along with an alert mechanism that delivers instant feedback to drivers. By doing so, the system will support smarter traffic enforcement, efficient resource management, and improved driver behavior in Sri Lanka.

2.1 Main Objective

The core objective of this study is to come up with a secure and user-friendly dashboard integrated with a driver alert system that enables traffic authorities to monitor overspending violations in real time, generate useful reports, and send immediate notifications to drivers to encourage safer driving practices.

2.2 Specific Objectives

- To create a secure web dashboard with login access restricted to authorized officers.
- To visualize real-time violations using tables, statistical charts, and heatmaps for better understanding.
- To add filtering functions such as by date, driver ID, or road location to make searching faster and easier.
- To provide export options (CSV and PDF) for reporting, record-keeping, and enforcement support.
- Designing a Driver Alert System that sends instant notifications such as SMS or email to drivers who commit speeding violations.

3. METHODOLOGY

3.1 Project Overview

The proposed project introduces an IoT and Machine Learning-based speed monitoring system with RFID-enabled licenses to improve traffic enforcement in Sri Lanka. Within this system, my contribution focuses on developing the Dashboard and Driver Alert System, which acts as the main interface for both monitoring and awareness.

The dashboard will serve as a central platform for traffic authorities, allowing them to log in securely and access real-time information about overspending violations collected through RFID checkpoints and IoT devices. The interface will display violations in an easy-to-understand format using tables and charts. In addition, the dashboard will include filtering options such as date, driver ID, and road location, enabling officers to quickly analyze specific cases. Authorities will also be able to generate exportable reports in formats such as CSV or PDF, supporting decision-making and official record-keeping.

The Driver Alert System will be integrated to provide direct communication with drivers. When a violation is detected, the system will send instant notifications such as SMS, reminding drivers of their offence. This real-time feedback is expected to increase driver awareness, discourage repeat violations, and promote safer road behavior.

By combining real-time visualization, reporting features, and direct driver engagement, this part of the project bridges the gap between enforcement and driver responsibility. It ensures that authorities gain actionable insights for resource planning while drivers are made aware of their behavior immediately, ultimately contributing to safer highways in Sri Lanka.

3.2. System diagram

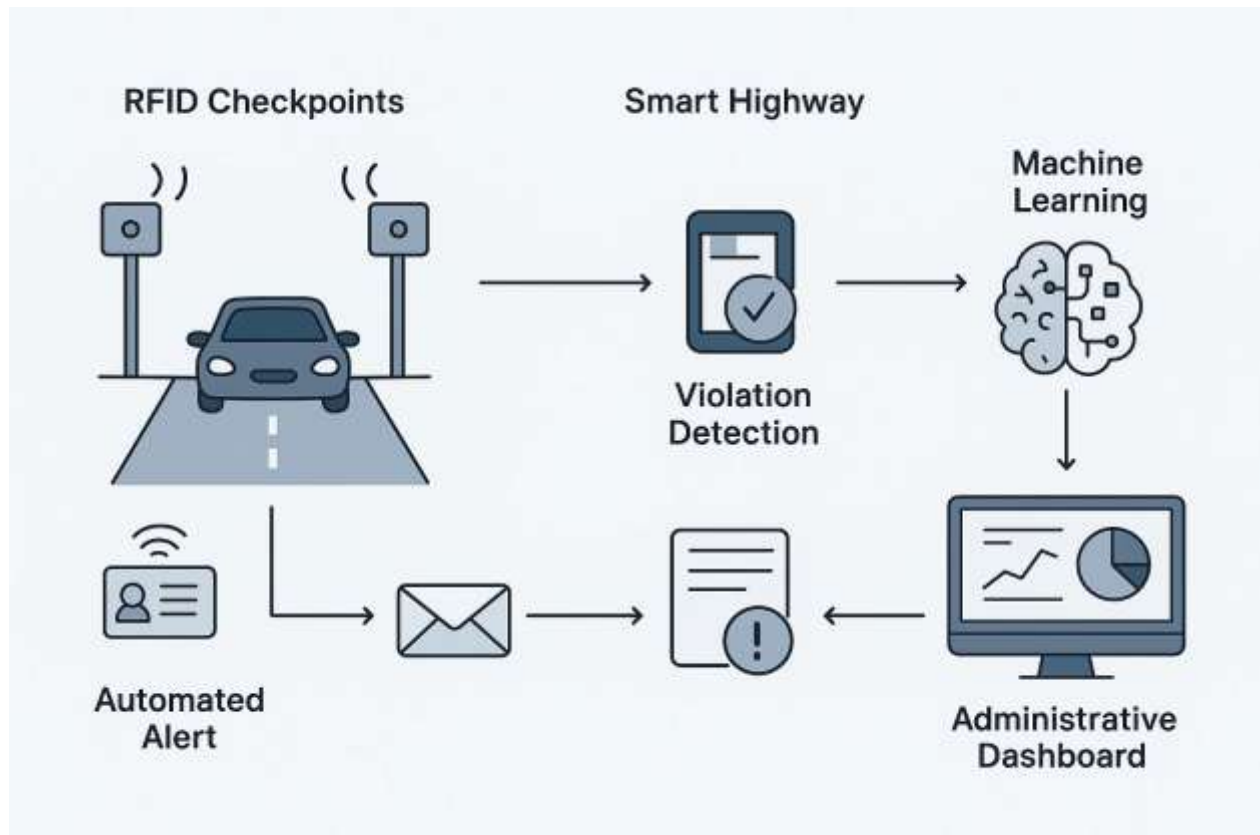


Figure 1 System diagram

3.2 Individual Component

3.2.1 Secure Web Dashboard

- Develop a secure web-based dashboard with login access restricted to traffic authorities.
- Provide a simple and user-friendly interface for monitoring overspending violations.
- Ensure that only authorized users can view and manage violation data.

3.2.2 Real-Time Data Visualization

- Display overspending violations in real time using tables, charts, and heatmaps.
- Highlight high-risk locations and time periods for better enforcement planning.
- Provide statistical summaries to support decision-making by authorities.

3.2.3 Filtering and Data Export

- Implement filtering functions to search for violations by date, driver ID, or location.
- Allow officers to generate official reports using export options in CSV and PDF.
- Provide easy access to both real-time and historical violation data.

3.2.4 Driver Alert System

- Notify drivers immediately of overspending violations through SMS or email alerts.
- Record each alert in the system for reference and analysis.
- Encourage driver awareness and reduce the chances of repeat offences.

3.2.5 Dashboard Integration with Backend

- Connect the dashboard with the backend database and APIs for real-time updates.
- Ensure smooth data flow from RFID checkpoints and IoT devices to the dashboard.
- Maintain data security and privacy while transferring sensitive driver information.

3.2.6 Support for Law Enforcement

- Provide centralized access for traffic authorities to monitor and analyze violations.
- Help officers plan resources more effectively by identifying trends and hotspots.
- Strengthen enforcement by combining real-time monitoring with predictive insights.

4. RESEARCH & DEVELOPMENT OVERVIEW

4.1 Sources for Test Data and Analysis

For the development of the dashboard and driver alert system, test data will mainly be collected from the simulated outputs of the RFID checkpoints and IoT devices developed by other group members. This data will include overspending records such as driver ID, vehicle type, location, time, and speed. In addition, synthetic or publicly available sample vehicle datasets may be used to test visualization and reporting features of the dashboard. The driver alert system will be tested using sample contact details linked with test IDs to check whether notifications are sent successfully.

4.2 Anticipated Benefits

The dashboard and driver alert system are expected to provide several benefits:

1. Real-time monitoring of overspending violations in a clear and user-friendly interface.
2. Improved decision-making for traffic authorities through charts, heatmaps, and statistical summaries.
3. Easy filtering and exporting of data for enforcement reports and policy planning.
4. Direct communication with drivers through instant notifications, increasing awareness and reducing repeat offences.
5. Better resource management for traffic officers by identifying high-risk times, areas, and drivers.

4.3 Scope and Specified Deliverables / Expected Research Outcome

The scope of this part of the project includes the design and implementation of:

- A secure web-based dashboard with login access for authorized users.
- Visualization tools to display real-time and predictive violation data.
- Filtering and exporting features for detailed analysis and reporting.
- A Driver Alert System that notifies drivers of violations through SMS or email.

Expected outcomes include a functional prototype dashboard that integrates with the backend system and successfully delivers real-time alerts to drivers. The system should enable both traffic authorities and drivers to actively participate in safer road management.

4.4 Research Constraints

The development of this part of the project may face several challenges, including:

- Limited access to real-world traffic violation data due to privacy concerns, requiring reliance on simulated datasets.
- Integration challenges between the dashboard, backend services, and IoT components.
- Possible costs involved in SMS or email notification services for driver alerts.
- Ensuring data security and privacy when handling sensitive driver information.
- Time limitations in implementing advanced visualization features such as heatmaps and predictive reports.

5. BUDGET AND BUDGET JUSTIFICATION

This research has been calculated to fit on its important parts such as hardware, software, and data collection and validation parts included in a carefully estimated budget. The allocation will provide cost-effectiveness in terms of achieving the technical needs of successful implementation and testing under real-world conditions in the Sri Lankan traffic conditions.

Item	Cost Estimate (LKR)	Justification
Cloud Computing Resources (AWS/Colab Pro)	12,000	For hosting, model training, and deployment
RFID Chip and Scanner	75,000	Required for license validation and speed data collection
Base64 or MIFARE Series	10,000	For secure data encoding and storage of driver information
Miscellaneous (Data Annotation Tools)	15,000	For labeling vehicle datasets and preparing training data
Internet and Contingency Expenses	10,000	To ensure continuous connectivity and system reliability
Chip Card Creation for ID	10,000	Each driver's ID card embedded with RFID chip

Figure 2 Budget plan

GANTT CHART

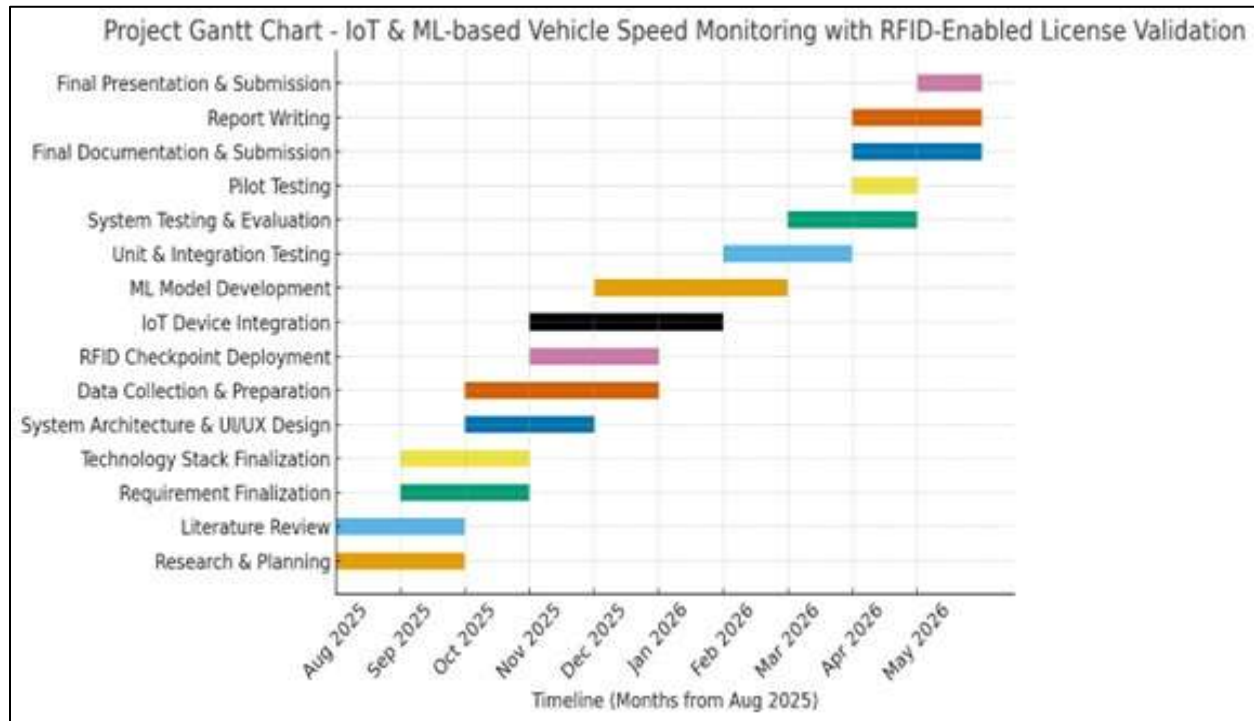


Figure 3 Gantt chart

WORK BREAKDOWN CHART

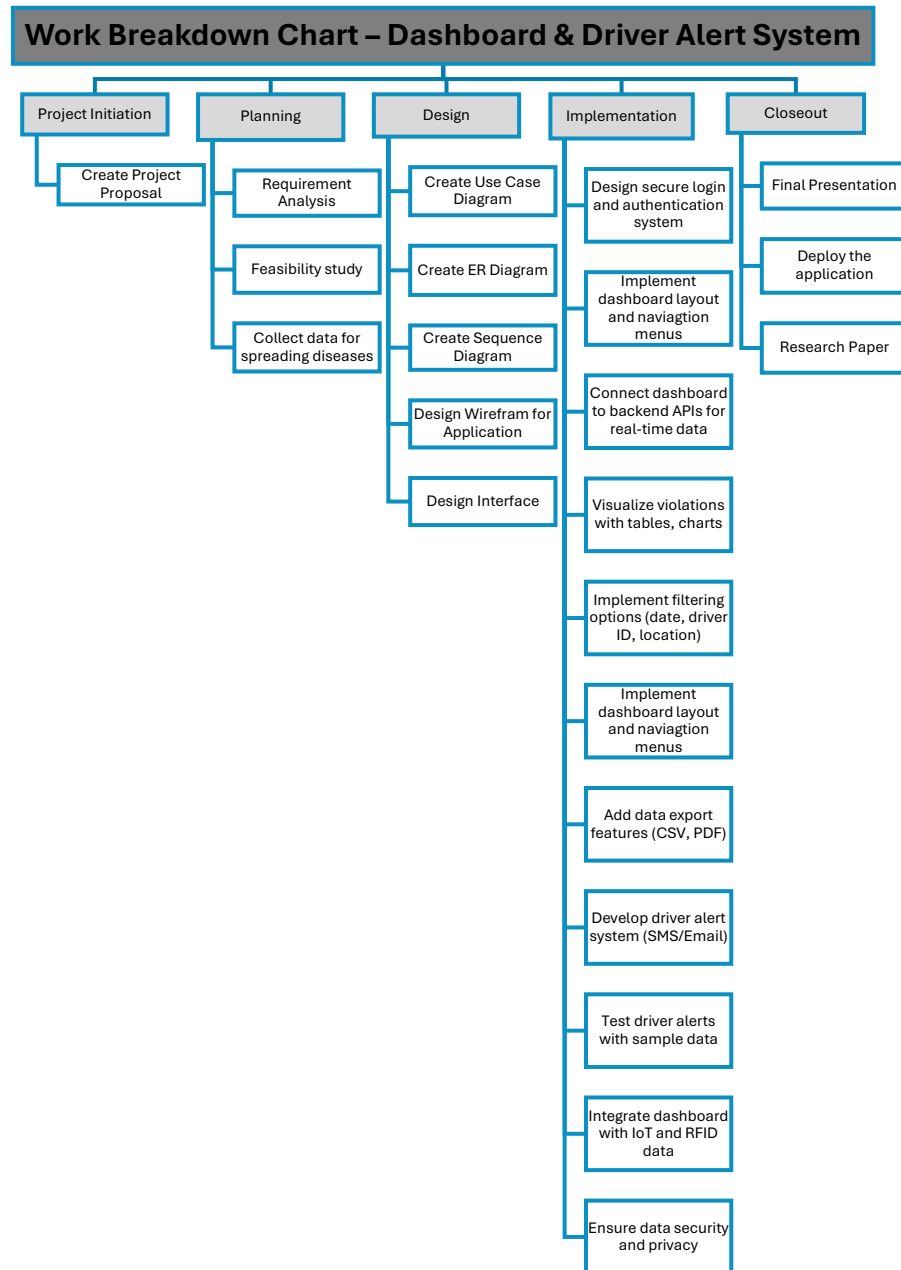


Figure 4 WBS diagram

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IOT & ML-BASED VEHICLE SPEED MONITORING WITH RFID-ENABLED LICENSE VALIDATION

Project ID: 25-26J-314

Project Proposal Report

K.M. Anuradha Keshani Konara

B.Sc. (Hons) Degree in Information Technology Specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

Backend Service for Violation Detection and Real-Time Alert Dispatch

Project ID: 25-26J-314

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Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Group Member Name	Student ID	Signature
Konara K.M.A. K	IT22350978	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Supervisor Signatures

Supervisor Name	Signature	Date
Dr. Kapila Dissanayaka		

ABSTRACT

Road traffic accidents remain a serious public safety concern in Sri Lanka, and over-speeding is also found to be one of the principal causes of death and injury. Although the issue is grave, there are not many rural and semi-urban areas that have basic automated enforcement setup since conventional camera, radar, and LiDAR-based sensing devices are very costly and involved in heavy maintenance charges. To fill this gap, this paper is intended for the design and implementation of a light-weight backend service to detect speed in real-time, secure data storage, and auto-routing of alert to enforcement agencies. The proposed backend collects speed data from RFID-based checkpoints through RESTful APIs, giving ensured and formatted communication between the server and IoT devices. Speed offense logic is used to check whether a vehicle has crossed over past allowable speeds in an area under surveillance. After confirming an offense, the system safely stores the incident information such as vehicle number, speed, time stamp, and location in a database like Firebase or MongoDB. To facilitate real-time enforcement, the backend is integrated with third-party communication services like Twilio or SendGrid, which send SMS or email alerts automatically to law enforcement authorities with all details. Communications are secured by utilizing database access control measures for the confidentiality and integrity of the sensitive data. With the integration of low-cost RFID technology, security, and the proposed backend service, the proposed backend service is a feasible and sustainable platform for the improvement of traffic law enforcement in Sri Lanka. The system minimizes human intervention, offers low-latency alerting and detection, and can be scaled up country-wide at low infrastructure costs. This work demonstrates how backend driven IoT solutions can significantly improve traffic monitoring, reduce accidents, and improve road safety outcomes in developing nations

Keywords— RFID, Speed Violation Detection, Backend Service, Real-Time Alerts, IoT, Traffic Enforcement

Table of Contents

DECLARATION	i
ABSTRACT	ii
List Of Figures	iv
List of Tables	iv
LIST OF ABBREVIATIONS	v
1 INTRODUCTION	1
1.1 Background & Literature Survey	2
1.2 Research Gap	4
1.3 Research Problem	6
2 OBJECTIVES	7
2.1 Main Objectives	7
2.2 Specific Objectives	7
3. METHODOLOGY	9
3.1 Overall System Description	9
3.2 Proposed Implementation Approach.....	9
3.5 Tasks and Sub-Tasks.....	13
3.6 Materials and Resources	14
4.PROJECT REQUIREMENTS	17
4.1 Functional Requirements.....	17
4.2 Non-Functional Requirements	17
4.3 User Requirements.....	17
4.4 System Requirements	18
4.4.1 Hardware Requirements.....	18
4.4.2 Software Requirements	18
4.4.3 Network Requirements	18
4.4.4 Security Requirements.....	18
4.4.5 System Integration Requirements	19
5.BUDGET AND BUDGET JUSTIFICATION	20

6. REFERENCES	21
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List Of Figures

Figure 1. Gantt Chart	11
Figure 2. Work Breakdown Chart	12
Figure 3. System Overview Diagram	15
Figure 4. System Architecture Diagram	16

List of Tables

Table 1. List of Abbreviation	v
Table 2. System Comparison.....	5
Table 3. Budget Plan	20

LIST OF ABBREVIATIONS

Table 1. List of Abbreviation

Abbreviations	Description
API	Application Programming Interface
DBMS	Database Management System
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
ML	Machine Learning
RFID	Radio Frequency Identification
SMS	Short Message Service
UI	User Interface

1 INTRODUCTION

The backend service is the central component of the system as far as the proposed traffic violation detection system is concerned, and it comprises the server-side logic that would be required for real-time processing of data, safe storage, and enforcement. It's primarily responsible for designing and developing a solid infrastructure that would efficiently process RFID-based vehicle speed data, enforce violation detection logic, and notify interested authorities instantaneously. With RESTful APIs, the backend enables seamless integration between RFID checkpoints over IoT and the database with zero latency, ensuring a glitch-free data flow. This allows for continuous monitoring across zones on multiple highways while supporting scalability for massive deployment.

One of the root functionalities of the backend is speed violation enforcement logic. After the violation is experienced, the system records the event in an encrypted cloud database, Firebase or MongoDB, thereby ensuring data integrity and supporting future analysis purposes. Besides, the backend synchronizes alert sending capabilities through integration with third-party APIs such as Twilio or SendGrid to transmit real-time SMS or email notification to local law enforcement units. The alerts are supplemented with critical details such as vehicle ID, location, speed, and timestamp to enable the enforcement officers to respond timely.

In addition to its utilitarian function, the backend emphasizes data confidentiality, safe communication, and system interoperability. Therefore, the backend service is the core of the project and enables efficient violation detection, safe processing of the data, and real-time assistance to law enforcement.

1.1 Background & Literature Survey

Road traffic accidents are a major global public safety issue, with over-speeding recognized as one of the primary causes of fatalities and serious injuries [1], [2]. In Sri Lanka, speed-related crashes account for a significant proportion of road deaths, particularly in rural and semi-urban areas where automated enforcement infrastructure is minimal [3], [4]. While developed countries widely employ Automatic Number Plate Recognition (ANPR) cameras, radar, and LiDAR-based enforcement, these technologies require high capital investment, complex installation, and continuous maintenance, making them unsuitable for large-scale deployment in low-resource settings [5], [6].

The emergence of the Internet of Things (IoT) has created opportunities for new approaches in Intelligent Transportation Systems (ITS). IoT enables real-time monitoring, data collection, and decision-making through distributed sensor networks [7]. Among these technologies, Radio Frequency Identification (RFID) has been successfully applied in toll collection, vehicle tracking, and access control due to its low cost, reliability, and tamper resistance [8], [9]. Integrating RFID with cloud computing platforms allows for processing and storing large volumes of traffic data with minimal on-site infrastructure [10].

Cloud service providers such as Firebase, AWS, and Azure offer serverless API hosting, real-time databases, authentication, and secure communication protocols [11], [12]. These capabilities support low-latency, scalable solutions that can seamlessly integrate with IoT devices for traffic enforcement applications. Previous research by Kumar et al. [13] demonstrated an RFID-based speed detection system that calculated average speed between checkpoints and issued alerts for violations. Similarly, Alavi et al. [14] implemented a cloud-connected speed monitoring system using ESP32 microcontrollers and MQTT/HTTP protocols to ensure efficient data transmission.

For violation notification, platforms such as Twilio and SendGrid are frequently employed to deliver SMS or email alerts to enforcement officers or vehicle owners

[15]. Other studies have explored AI and machine learning techniques to predict accident-prone zones and identify repeat offenders [16], [17], contributing to proactive enforcement strategies.

Despite these advancements, most existing systems are hardware-intensive, rely on high-maintenance sensors or cameras, and lack the cost-effectiveness and scalability required for developing countries [18], [19]. Furthermore, many implementations fail to address digital security risks when transmitting sensitive driver and vehicle data [20].

1.2 Research Gap

A thorough review of existing literature reveals that while numerous systems have been developed for vehicle speed detection and traffic violation enforcement using radar, LiDAR, and camera-based ANPR technologies, there is limited focus on low-cost, scalable solutions suitable for developing countries such as Sri Lanka. Most existing deployments are hardware-intensive, require expensive maintenance, and rely on complex installation processes, making them impractical for wide-scale adoption in low-resource settings.

Research A [13] proposed an RFID-based vehicle speed detection mechanism that calculates average speed between two checkpoints and triggers violation alerts when thresholds are exceeded. However, the system was implemented as a standalone prototype with no cloud integration, limiting its scalability and real-time enforcement capabilities. Additionally, there was no provision for secure remote access, automated officer notifications, or centralized violation databases.

Research B [14] demonstrated a cloud-connected speed monitoring system using ESP32 microcontrollers with MQTT/HTTP protocols for data transmission. While the system achieved efficient communication, it lacked advanced zone-specific violation logic and did not address data privacy concerns during transmission. Furthermore, it depended on localized server infrastructure, which may cause latency issues in high-volume scenarios.

Research C [15] focused on AI-based accident prediction models that incorporated speed data and historical violation patterns. Although this research demonstrated predictive capabilities, it did not integrate a practical, low-cost field-level speed monitoring hardware solution. The emphasis remained on post-event analysis rather than real-time violation detection and alerting.

Table 2. System Comparison

Features	A	B	C	Proposed System
RFID-based detection	✓	✗	✗	✓
Speed violation logic	✗	✗	✗	✓
Automated SMS/Email alerts	✗	✗	✗	✓
Secure data transfer	✗	✗	✗	✓
Low-cost scalability	✓	✓	✗	✓

Traditional camera and radar-based enforcement methods remain unsuitable for rural and semi-urban Sri Lanka due to their expense of operation, environmental sensitivity, and maintenance needs. Without an inexpensive, secure, and scalable system, much of the land remains unmonitored, and habitual speed offenses remain unpunished.

The suggested study addresses these requirements with a lightweight, secure, RFID + backend service for real-time speed violation detection. The system will gather RFID checkpoint information using RESTful APIs, apply dynamic violation rules for different road zones, store records safely in a database, and provide automatic notification to enforcement officers via SMS or email.

This project fills the void for a holistic, secure, and cost-effective RFID-based speed enforcement solution tailored to the infrastructure and budget limitations of Sri Lanka. The proposed system will utilize optimally positioned RFID checkpoints to collect vehicle speed data and transmit it securely to a centralized backend in the form of RESTful APIs. It will utilize a dynamic, zone-based violation detection method with all violation records stored in a secure, centralized database. Enforcement officers will receive automated real-time alerts via SMS and email by services such as Twilio or SendGrid. The system will be low cost in terms of deployment and maintenance and scalable for country-wide roll out.

1.3 Research Problem

Traffic accidents on roads remain a public safety concern in Sri Lanka, and over-speeding has been among the principal contributors to casualties and major injuries. Statistical reports issued by the Sri Lanka Police indicate that high proportions of speed-related accidents occur in rural and semi-urban areas where there are no permanent speed-checking facilities. Modern enforcement technology such as radar guns, LiDAR, and ANPR camera systems not only involve huge costs but require trained personnel to operate and maintain constantly, making them unsuitable for mass deployment. Therefore, much of the road network is under-monitored, creating scope for habitual offenders to drive recklessly without risking detection.

In rural areas, enforcement officials must rely on mobile patrols or makeshift checkpoints. Semi-urban areas have seen pilot deployments of RFID and IoT-enabled speed detection prototype systems. But these usually operate in stand-alone mode, storing data locally on the device. Without integration with a centralized backend or automated alerting systems, offenses tend to remain undetected by enforcement officials and do not result in timely punishment or aid repeat offender tracking. These limitations detract from the overall process of enforcement and diminish its ability to deter unsafe driving habits.

Furthermore, most existing systems lack safe, real-time data transmission mechanisms, exposing sensitive vehicle and driver information to cybersecurity risks. Therefore, there is an immediate need for a low-cost, secure, and scalable backend service that not only detects speed offenses in real time but also safely stores data and transmits instant alerts to law enforcement. The study fills this void by designing an RFID-based backend system that can impose zone-specific violation logic, securely log violation events, and automatically alert involved authorities via SMS or email. This method assures detection events always result in enforcement action without being overly expensive or scalable for use in the unmonitored areas of Sri Lanka.

2 OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design and develop a secure, low-cost RFID-based speed enforcement system that can detect zone-specific violations and automatically dispatch real-time alerts to law enforcement authorities. The system is specifically tailored to Sri Lanka's infrastructure and budget limitations, making it both practical and scalable. By integrating RESTful APIs, secure databases and secure communication channels such as SMS or email, this backend service ensures timely enforcement and enhances overall road safety through efficient monitoring and reliable violation tracking.

2.2 Specific Objectives

1. To implement RFID checkpoints for capturing and transmitting vehicle speed data via secure RESTful APIs

This ensures that vehicles are automatically tracked at entry and exit points, and their speed is calculated accurately. Using secure APIs allows safe and real-time data transfer to the backend system.

2. To design and integrate a dynamic, speed violation detection algorithm for accurate speed monitoring

Different road zones (highways, rural roads, school zones, etc.) have different speed limits. This algorithm will compare vehicle speeds against those zone-specific limits and identify violations precisely.

3. To store violation records in a secure, centralized database with controlled access for authorized personnel

All violation data will be stored in one reliable cloud database (Firestore/MongoDB), protected by access control, so only enforcement officers can view or manage the records.

4. To develop an automated alert mechanism for sending real-time SMS and email notifications to enforcement officers using services such as Twilio or SendGrid

As soon as a violation is detected, the system will automatically notify the nearest law enforcement officials, including details like vehicle ID, location, time, and speed, without delays or manual intervention.

5. To optimize the system for minimal deployment and maintenance costs, ensuring nationwide scalability

The system will be designed to work with affordable IoT/RFID hardware making it cost-effective to install and maintain across many regions in Sri Lanka.

3. METHODOLOGY

3.1 Overall System Description

The backend methodology involves developing RESTful APIs to collect RFID speed data, implementing speed violation logic, securely storing records in databases like Firebase or MongoDB, and integrating real-time alert systems via SMS or email. This enables automated, scalable traffic enforcement with minimal human intervention.

3.2 Proposed Implementation Approach

The development will follow an iterative approach, consisting of the following phases:

Step 1: Requirement Gathering

- Identify system requirements for handling RFID-based speed data, violation detection, secure storage, and real-time alerts.
- Consult traffic law enforcement authorities to validate key operational and legal needs.

Step 2: Data Handling & Preprocessing

- Collect and preprocess vehicle speed datasets from RFID checkpoints.
- Normalize and validate data to reduce errors and false detections.
- Define thresholds for different road zones.

Step 3: Backend Development

- Implement RESTful APIs for receiving and processing RFID speed data.
- Apply road zone-specific violation detection logic.
- Store violation records securely using databases (e.g., Firebase/MongoDB).

Step 4: Testing and Validation

- Test system performance for low-latency data processing and alert accuracy.
- Validate violation detection against real-world datasets.
- Conduct pilot testing with simulated traffic scenarios.

Step 5: Deployment

- Deploy the backend as a scalable cloud-based service.
- Integrate external APIs (e.g., Twilio, SendGrid) for SMS/email alerts.
- Provide real-time violation reporting dashboards for law enforcement authorities.

3.3 Gantt Chart

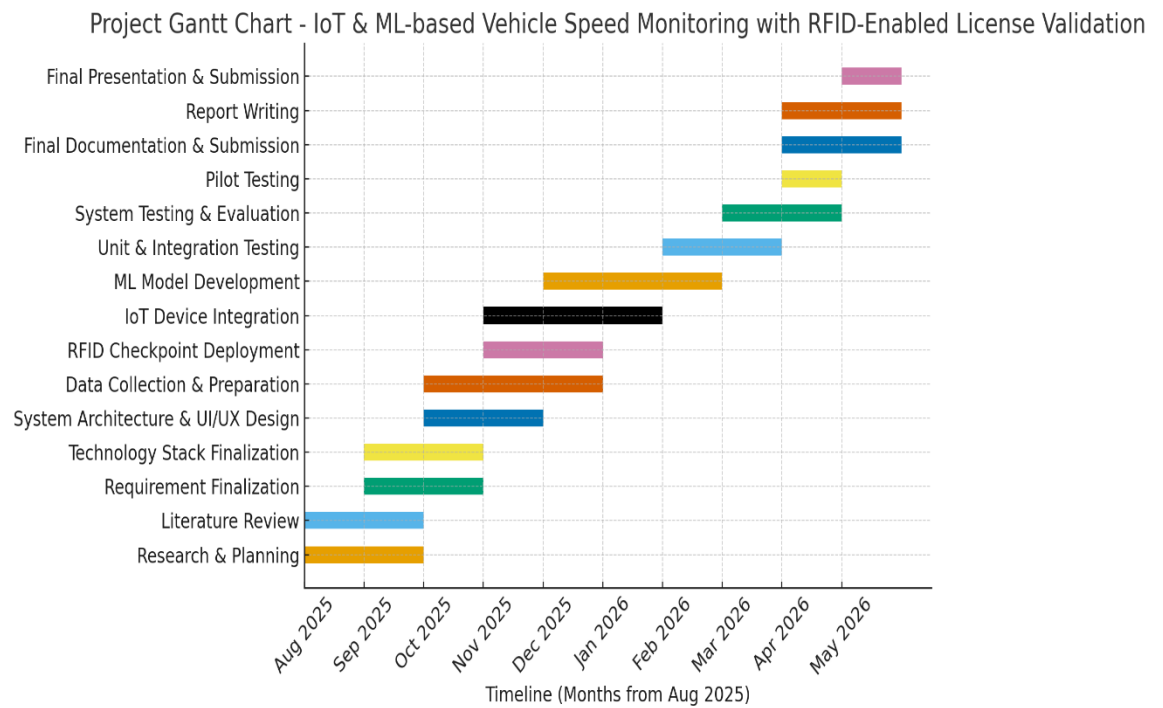


Figure 1. Gantt Chart

3.4 Work Breakdown Structure

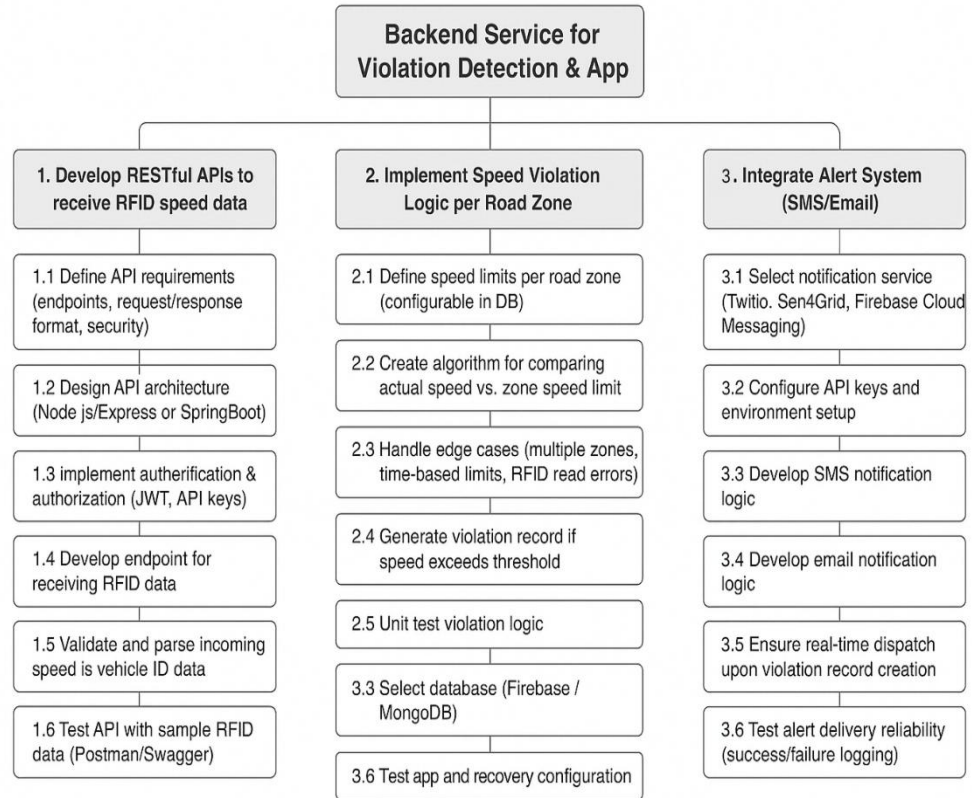


Figure 2. Work Breakdown Chart

3.5 Tasks and Sub-Tasks

1. API Development

- Design and develop RESTful APIs to receive RFID speed data from checkpoints.
- Implement authentication and encryption for secure data transmission.

2. Violation Detection Logic

- Implement dynamic, zone-specific speed limit configurations.
- Calculate vehicle speed from checkpoint timestamps and compare against set limits.

3. Data Storage

- Integrate Firebase or MongoDB as a centralized, secure database.
- Implement access controls to restrict violation record access to authorized personnel.

4. Real-Time Alerts

- Integrate Twilio/SendGrid APIs for SMS and email alerts.
- Configure alert templates with vehicle details, location, and violation time.

3.6 Materials and Resources

- 1. Software:** Node.js, Firebase/MongoDB, RESTful API framework, Twilio/SendGrid for alerts.
- 2. Hardware:** Development laptop / cloud deployment environment (AWS, GCP, or Azure).

3. Data Collection

- **Quantitative:** Vehicle speed data from RFID checkpoints, road zone speed limits, violation history logs.
- **Qualitative:** Expert feedback from traffic police on accuracy of alerts and violation reports.

4. Timeframe

This component is expected to be completed in 3 months:

- **Month 1:** Requirement gathering (discussions with law enforcement, identification of system needs).
- **Month 2:** Backend development (RESTful APIs for data reception, violation logic per road zone).
- **Month 3:** Database integration (Firebase/MongoDB) and alert system implementation (SMS/email).
- **Month 4:** System testing (low-latency processing, validation with real datasets).
- **Month 5:** Pilot deployment in selected checkpoints, performance tuning, feedback collection.
- **Month 6:** Final deployment, cloud integration, and preparation of evaluation report.

5. Expected Outcome

- Reliable and automated speed violation detection system.
- Real-time alerts to law enforcement with minimal human intervention.
- Scalable, transparent, and secure backend service for traffic monitoring.

3.7 Overall System Architecture Diagrams

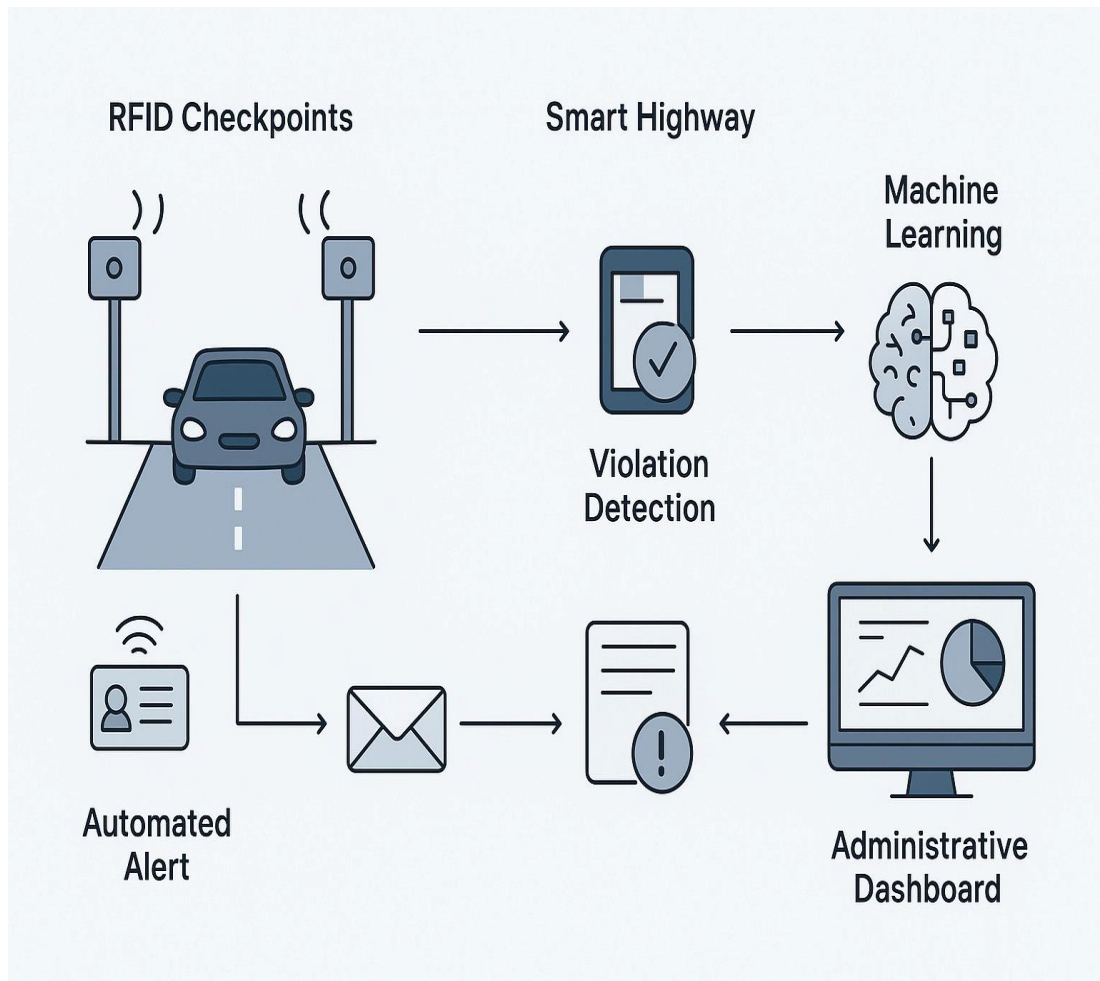


Figure 3. System Overview Diagram

3.8 System architecture of the backend service for violation detection and real-time alert dispatch

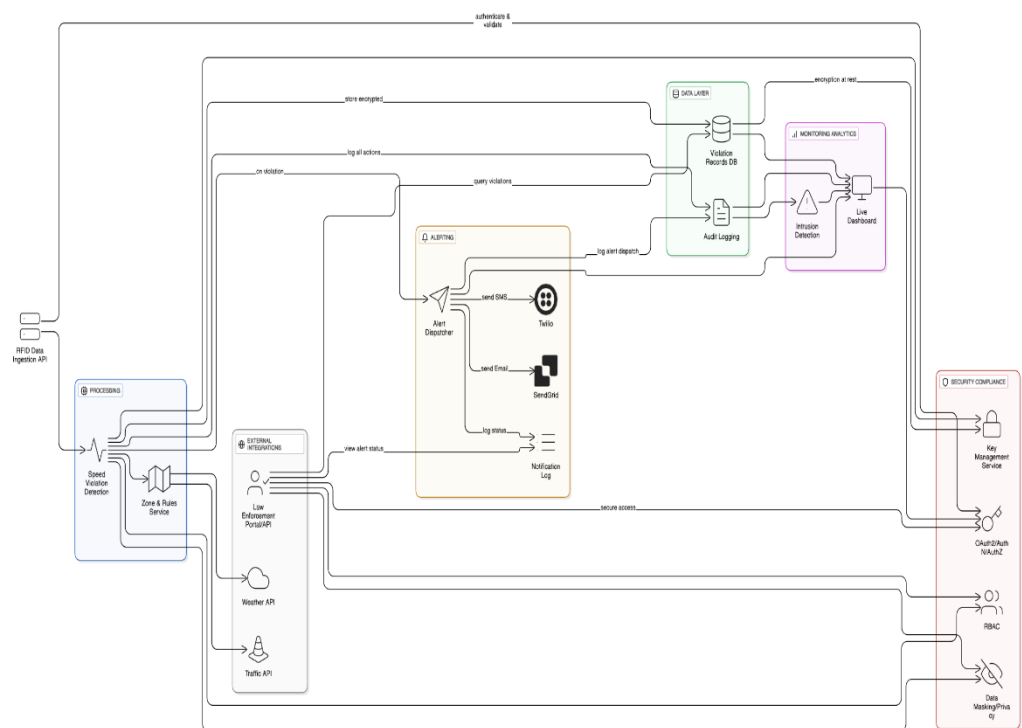


Figure 4. System Architecture Diagram

4.PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system shall receive RFID data through RESTful APIs.
- The system should calculate vehicle speed and compare it with the zone-specific speed limit.
- The system shall detect violations if the speed exceeds the defined limit.
- The system shall store violation details (RFID ID, vehicle number, timestamp, speed, zone, violation status) in a secure database.
- The system shall send real-time alerts via SMS/email to law enforcement officers upon violation detection.

4.2 Non-Functional Requirements

- Performance: The system should process and store data within 2 seconds of receiving RFID input.
- Scalability: The system should support multiple RFID checkpoints and thousands of concurrent records.
- Security: Data should be encrypted during transmission (HTTPS) and storage (database security rules).
- Maintainability: Codebase should follow modular design to allow easy updates.
- Usability: Alerts should be clear, concise, and in a format that officers can quickly understand.
- Interoperability: The system should be integrated with external services (Twilio, Firebase, law enforcement dashboards).

4.3 User Requirements

- The system should automatically detect speed violations without manual intervention.
- Law enforcement officers should receive real-time alerts (via SMS/email) when violations occur
- Administrators should be able to access logs and reports of violations.

- The system should work continuously with high reliability and minimal downtime.

4.4 System Requirements

4.4.1 Hardware Requirements

RFID checkpoints consist of RFID readers installed at road checkpoints and RFID tags carried by vehicles, enabling the system to automatically read vehicle speed and identification details. Police personnel can see this on their mobile phones or laptops, receiving real-time alerts via SMS or email for every offense detected. RFID checkpoints involve RFID readers installed at checkpoints along roads and RFID tags mounted on automobiles, so the system can automatically capture car speed and ID data. Police officers have access to such information through mobile phones or computers, receiving real-time signals through SMS or email for any offenses detected.

4.4.2 Software Requirements

The backend of the system is coded with RESTful APIs for RFID data input handling and alert sending. Data storage in real-time is handled by Firebase or MongoDB to ensure efficient and secure records of all vehicle violations. Twilio offers law enforcement notifications for SMS notifications and SendGrid for email notifications, enabling quick responses. Development is made easier through version control platforms such as GitHub or GitLab so that there may be seamless collaboration and effective upkeep of the project codebase.

4.4.3 Network Requirements

The system ensures safe communication among modules through HTTPS from RFID stations to the back-end server. In addition, reliable network connectivity is ensured for smooth transmission of data to identify vehicle speed violations accurately and in time.

4.4.4 Security Requirements

All data communicated between RFID checkpoints, and the backend server is encrypted to ensure confidentiality and prevent unauthorized access.

4.4.5 System Integration Requirements

The backend application is coded in a way that it can seamlessly integrate with RFID data sources to enable real-time vehicle speed calculation and violation detection. The system is further integrated with notification services such as Twilio and SendGrid for sending SMS and email notifications to law enforcement personnel

5.BUDGET AND BUDGET JUSTIFICATION

This research has been calculated to fit on its important parts such as hardware, software, and data collection and validation parts included in a carefully estimated budget. The allocation will provide cost-effectiveness in terms of achieving the technical needs of successful implementation and testing under real-world conditions in the Sri Lankan traffic conditions.

Table 3. Budget Plan

Item	Cost Estimate (LKR)	Justification
Cloud Computing Resources (AWS/Colab Pro)	12,000	For hosting, model training, and deployment
RFID Chip and Scanner	75,000	Required for license validation and speed data collection
Base64 or MIFARE Series	10,000	For secure data encoding and storage of driver information
Miscellaneous (Data Annotation Tools)	15,000	For labeling vehicle datasets and preparing training data
Internet and Contingency Expenses	10,000	To ensure continuous connectivity and system reliability
Chip Card Creation for ID	10,000	Each driver's ID card embedded with RFID chip

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